Imagining Human-Robot Future Homes: A Feminist HRI Approach through Speculative Design in South Korea

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How might Human-Robot Interaction (HRI) challenge social norms in societies where feminist discourse is politically sensitive? This study explores that question through two participatory speculative design workshops conducted in South Korea, a context marked by rapid technological development and contested gender politics. Drawing on Feminist HRI and post-phenomenology, the study designed a speculative "what-if" scenario in which every household must adopt a robot under a fictional government policy. Through the workshops, 19 participants subtly critiqued emotional labor, power hierarchies, and family roles, expressing what can be interpreted as "quiet feminism." The resulting speculative artifacts, including a collaboratively created Future Robot Home Catalog, revealed ambivalent imaginaries: robots were envisioned as both helpful and potentially alienating, norm-reinforcing and socially transformative. The study finds that speculative design serves not only as a method of imagination but also as a feminist practice, enabling culturally situated critique, ethical reflection, and the articulation of plural, relational futures. These findings contribute to the growing field of Feminist HRI by demonstrating how speculative design can serve as a culturally sensitive and inclusive method for navigating sociotechnical complexities and supporting inclusive design practices beyond universal paradigms.

CCS CONCEPTS • Human-centered computing • Interaction design • Interaction design process and methods

Additional Keywords and Phrases: Participatory Design, Speculative Design, Feminist HRI, Sociotechnical Imaginaries, Human-Robot Interaction

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1 INTRODUCTION

Recent research on Feminist Human-Robot Interaction (Feminist HRI) has sought to critically examine the underlying structures in the current HRI field, which has traditionally prioritized usability, efficiency, or acceptance, highlighting how robotic systems are shaped by cultural expectations, power dynamics, and existing social norms [7, 82]. However, much of this work has emerged from Western contexts where feminist discourse is more publicly accepted [7, 27, 82], leaving non-Western societies underexplored within this framework. Given the sociocultural and political complexities surrounding gender discourse in these regions [53, 83], there is a need to investigate how Feminist HRI principles might manifest in non-Western cultural and technological landscapes.

South Korea presents a particularly compelling case for this due to the country's contradictory relationship with technology and gender norms. While the country embraces technological innovation, it simultaneously struggles with the politicization of gender discourse, persistent gender inequality [38, 54], and a politically charged backlash against feminism [30, 40, 46]. Within this landscape, domestic robots offer a rich site of inquiry. As technologies, such as robots and AI, increasingly enter private and emotionally charged spaces, including the home, they bring with them not just functionality but new dynamics of authority, caregiving, and relational labor [72]. The challenge, then, is not only to examine what these robots can do but to ask what kinds of domestic futures they are imagined to support and by whom.

This study investigates how Human-Robot Interaction (HRI) can serve as a site for imagining social and domestic norms in South Korea, particularly those related to family structure, cultural values, and responsibilities. Through two participatory speculative design workshops, it explores how individuals collectively imagine alternative robotic futures and how these sociotechnical imaginaries reflect, resist, or reconfigure dominant narratives about technology and home. Drawing on Feminist HRI and post-phenomenology, the research introduces a culturally sensitive framework grounded in indirect expression, which has been described as "quiet feminism" [40]. This study also seeks to contribute to the emerging conversations in Feminist HRI by offering a culturally situated perspective and demonstrating how speculative design methods can foreground cultural specificity, pluralism, and inclusion [80, 82].

1.1 Research Question(s)

The following research question guides this study:

"How can Human-Robot Interaction (HRI) in South Korea challenge social norms and reconfigure future domestic spaces?"

To answer this, this research explores the following sub-research questions: "In what ways do existing gender roles, labor expectations, and family structures influence perceptions of domestic robotics?" and "What sociotechnical imaginaries of robots in domestic spaces emerge through participatory speculative design?"

2 BACKGROUND

2.1 Human-Robot Interaction (HRI) and Shifting Family Structures in South Korea

South Korea is internationally recognized as one of the leading countries in technological innovation, particularly in robotics, artificial intelligence (AI), and 5G networks [8, 60]. Robotics research and development, in particular, have been heavily promoted by government initiatives and private sector investments, positioning the country at the forefront of human-robot interaction (HRI) research [61]. Robots are now a familiar sight in Korean airports [49], museums [5], and restaurants [16]. Domestically, service robots for caregiving and household assistance are increasingly present, creating new forms of intimacy and labor within everyday life [51].

Simultaneously, South Korea faces profound demographic and societal shifts. With one of the world's lowest birth rates, a rapidly aging population, and an increasing number of single-person and unmarried households ("Bi-hon"), traditional family structures are being fundamentally reconfigured [2, 42, 43]. These demographic transformations have spurred the development of technologies such as domestic robotics, offering new possibilities for addressing emerging gaps in caregiving, cohabitation, and familial responsibility, intensifying public discourse around the future of domestic life [41].

The intersection of these technological advancements and demographic transformations opens new discursive spaces for inquiry. As robotics becomes increasingly capable of performing not only physical tasks but also emotional and caregiving roles, the question arises: How might the integration of robots into homes reshape concepts of family, care,

labor, and social norms? While the current deployment of robots remains relatively limited to functional tasks, the evolving technical landscape suggests that more profound societal impacts are possible [56].

This study, therefore, takes these tensions as a starting point to employ a speculative approach. Rather than assuming a linear adoption of technologies, it seeks to explore how future imaginaries of technology (robots) could challenge or reinforce existing domestic norms in South Korea. The specific speculative scenario, developed to critically engage with these possibilities, will be introduced in 3.1.1 Design Fiction.

2.2 "Feminism" in South Korea

In South Korea, the word "feminism" itself carries a heavy and often negative connotation. Although feminist movements have gained greater visibility in recent years, public attitudes toward feminism remain deeply polarized [46]. The mainstream media, online communities, and political figures often frame feminism as a source of social division rather than as a struggle for gender equality [38, 48]. This stigmatization contributes to widespread backlash, not only against feminist activism but also against individuals perceived as sympathetic to feminist causes, including those in professional and educational environments [17].

The backlash has significant implications for academic, public, and political discourse. Academic research addressing gender inequality, feminist theory, or gender-based violence frequently encounters institutional resistance, limited funding, or public scrutiny [46]. Public discussions of feminism are often depoliticized or reframed under more "neutral" terms, such as "equality" or "diversity," sidestepping direct engagement with feminist critiques [30, 38].

Within this constrained environment, some individuals have turned to what scholars describe as "quiet feminism" [40]. Rather than engaging in overt political activism, quiet feminists integrate feminist principles into their everyday lives, subtly practicing *everyday feminism* and challenging traditional gender norms without explicitly identifying their actions as feminist [40, 69].

Understanding this cultural and political backdrop is crucial for examining how feminist values might emerge or be suppressed in speculative exercises about future domestic technologies. It also highlights the importance of creating participatory design spaces that allow for indirect, nuanced expressions of feminist critique, particularly in a context where overt feminism may be fraught.

2.3 Feminist HRI

Feminist HRI (Human-Robot Interaction) provides a critical lens for examining how technologies are conceptualized, designed, and embedded within social structures, while "feminism" often connotes broad social movements advocating for gender equality [6, 7, 82]. This approach offers a more focused methodological framework that views technology as inherently political and entangled with social norms, power relations, and structural exclusions [76].

Central to this framework are five guiding principles outlined in recent scholarship: participation and making labor visible, pluralism, elevating emotion and embodiment, re-thinking binaries and hierarchies, and self-disclosure, which collectively emphasize inclusive design, situated perspectives, affective and embodied experience, the disruption of normative power structures, and critical reflection [82].

However, despite its growing presence in academic discourse, much of the existing Feminist HRI literature has been developed in Western contexts such as Sweden, Germany, and the United States, where feminist discourse is more publicly accepted or at least openly debated [7, 27, 82]. In contrast, in non-Western societies such as South Korea, feminist language can be highly politicized, socially stigmatized, or even perceived as confrontational or insulting [30,

31]. This cultural context complicates the direct application of openly feminist frameworks like Feminist HCI/HRI, highlighting the need for more contextually sensitive approaches.

This study addresses this gap by integrating speculative design with Feminist HRI. Rather than seeking universal Feminist HRI approaches, it foregrounds how relational, embodied, and culturally embedded practices can expand the field of Feminist HRI toward more context-sensitive, pluralistic directions. The study does not aim to provoke or expose gender tensions, but to gently ease them by creating space for indirect, safe, and culturally resonant conversations about care, family, and future technologies.

While framed through HRI due to its focus on domestic robots, it does not draw a strict distinction between HRI and HCI. Following prior scholarship, it treats them as overlapping domains within the broader human-technology interaction, extending conceptual and methodological tools across both fields [32, 82].

2.4 Sociotechnical Imaginaries

Technical developments are always entangled with collective visions of social futures [65]. Sociotechnological imaginaries refer to these collectively held and performed visions of desirable futures (or of resistance against the undesirable), animated by shared understandings of forms of social life and social order attained through and supportive of advances in science and technology [37]. These imaginaries not only guide what technologies are developed but also shape a shared understanding of what forms of life are possible, acceptable, or ideal [52].

In robotics, social robots are often considered as solutions to broader societal challenges, such as aging populations [68, 78], labor shortage [58], or educational reform [15]. However, sociotechnical imaginaries do not merely reflect technical possibilities; they articulate cultural values, ethical concerns, and political hopes [55]. Through the lens of mutual shaping, technology and society are seen as co-constructive, as social and cultural factors influence the design and use of robots, while robotic technologies, in turn, mediate constructions of social values and meanings [67].

Accordingly, this study uses sociotechnical imaginaries not just as a descriptive concept but as a critical framework for interpreting participants' speculative robot designs. These designs are understood as situated acts of imagination, expressions of deeply embedded collective hopes, anxieties, and contestations about future domestic life. Through analyzing these artifacts, the study seeks to surface the often-unspoken values that shape what kinds of futures are seen as thinkable, desirable, or attainable [55, 74].

Sociotechnical imaginaries link directly to post-phenomenology, elaborated in the following section, which similarly emphasizes the co-constitution of humans, technologies, and the world [74]. Together, these frameworks enable this study to explore not only individual experiences but also the unique sociocultural narratives shaping the human-robot domestic future in South Korea.

2.5 Theoretical Framework: Post-phenomenology

Post-phenomenology, a philosophical framework developed most prominently by Don Ihde, provides a lens for examining how technologies, including robots, are not merely passive tools but active mediators that transform human routines, values, perceptions, and relationships [34, 35]. Departing from the classical phenomenology of thinkers like Husserl and Merleau-Ponty, it challenges purely human-centered views [36]. While traditional phenomenology described how humans experience the world through consciousness or embodied being, it often treats technologies as background or secondary [33, 57]. In contrast, post-phenomenology highlights technologies as co-constitutive actors: humans and technologies mutually shape each other within every experience [26, 34, 77].

This framework introduces two major shifts. First, technologies are seen as actively shaping both perception and action rather than simply extending human capabilities or withdrawing from attention when functioning properly [34]. Second, agency is no longer conceived as exclusively human; it is distributed across human-technology assemblages [26]. Consequently, designers are engaged in embedding ethical, social, and behavioral scripts into technological artifacts, making technological mediation an unavoidable part of life [77].

In the field of HRI, post-phenomenology redirects attention from robots' technical functions to their roles in reconfiguring human-world relations [9]. Here, robots are not simply assistants or social actors but play a role in transforming cultural values, norms, and social structures. For example, robots designed to support productivity can reshape human routines and expectations [74], and robots designed for caregiving mediate spatial arrangements, emotional atmosphere, and norms of dependency and interaction [62]. In this sense, robots become sites where cultural values, ethical concerns, and relational norms are materially enacted.

Applying a post-phenomenological perspective is crucial for this study. Participants' speculative robot designs are not interpreted merely as creative outputs but as expressions of mediated domestic imaginaries, revealing how technologies might reconfigure social norms around care, labor, and family in South Korea. Post-phenomenology enables this study to move beyond utility-based or anthropomorphic views of robots and how imagined robots participate in constituting new domestic realities. It acknowledges that human-robot relations are not fixed but emergent, relational, and culturally situated [79]. Thus, this offers a critical foundation for analyzing the deeper cultural and ethical dimensions of speculative HRI futures.

3 METHODOLOGY

3.1 Participatory Speculative Design

This study follows a qualitative and critical methodology, integrating speculative design with participatory design approaches. As conceptualized by Dunne and Raby, speculative design is not concerned with predicting the future but with probing possibilities, questioning the status quo, and opening up discursive spaces about what kinds of futures are desirable or undesirable [20]. In this study, the speculative approach is not used as a problem-solving tool but as a method to create artifacts, scenarios, or narratives that provoke critical reflection [25].

A participatory approach was integrated to invite diverse voices into the speculative design process [6]. This approach enabled non-designers to collaboratively envision alternative robotic futures without reducing these futures to purely technical functionalities [22], which also aligns with feminist design values of pluralism and inclusion [6, 82]. It is important to note that although Feminist HCI/HRI principles informed the whole workshop framework, the term "feminism" was not explicitly mentioned during the sessions. Instead, feminist critiques and perspectives will be embedded into the analytical and interpretive layers of the research [7].

3.1.1 Design Fiction

To operationalize the speculative design engagement, the workshops were framed around a central "what-if" scenario: *"What if all families (in South Korea) have to adopt at least one robot in their home?"* [29] The fictional legislative context, the "Household Robotics Act," was designed in response to ongoing shifts in Korean family structures, including the rise of single-person households, declining birth rates, and an aging society [59]. The scenario was designed to foreground social, ethical, and relational issues, including shifting notions of kinship, domestic labor, and emotional care. Thus, the

workshops became a space to question existing social norms and imagine how domestic life might be reconfigured through mandated human-robot cohabitation.

A short pre-study was conducted prior to the workshops to iteratively test the speculative scenario and activity format, which informed refinements to the workshop format, structure, and design canvas. Participants engaged in two workshops that were built sequentially. The first workshop emphasized the rapid, individual exploration of speculative futures through brainstorming and structured world-building. Then the second structured these ideas into a collaborative design fiction artifact [70]**:** The Future Robot Home Catalog. The catalog format was chosen for its familiarity and narrative flexibility, inspired by speculative design practices like the "Future IKEA Catalog" from Near Future Laboratory and the Mobile Life Centre [10, 14]. This allows participants to translate abstract ideas into materialized and tangible everyday imaginaries of future domestic life that serve as a prompt for ongoing dialogue and exploration [11].

3.2 Data Collection

This study was conducted through two participatory speculative design workshops held in March 2025 at Seoul National University. A total of 19 individual participants were engaged across both workshops. 14 participants attended Workshop 1, and 10 participants attended Workshop 2, with half of the participants in Workshop 1 also participating in Workshop 2. Participants were primarily graduate students and researchers with academic backgrounds spanning the social sciences, humanities, design, and technology-related disciplines, reflecting diverse perspectives relevant to Human-Robot Interaction (HRI) and speculative design.

Recruitment was carried out through university networks and internal invitations. All participants were asked to complete a pre-questionnaire assessing demographics, prior experience, and familiarity with speculative design, robotics, and HRI before participation. The majority (16 out of 19 participants, 84%) identified as female, with 3 participants identifying as male. Most participants were aged 25–29 years (11), followed by participants aged 19-24 years (4), 30-34 years (3), and 45+ years (1). Notably, 14 participants (74%) reported no prior experience with speculative design, and only one participant had attended another speculative design workshop before. Most participants rated themselves between 2 and 4 on a 5-point scale regarding familiarity with robotics and HRI, indicating a range of awareness levels without a dominant expert perspective. (The full pre-questionnaire can be found in Appendix A.1, and participant demographic details in Appendix A.2.)



Figure 1: Workshop 1 in the classroom setting

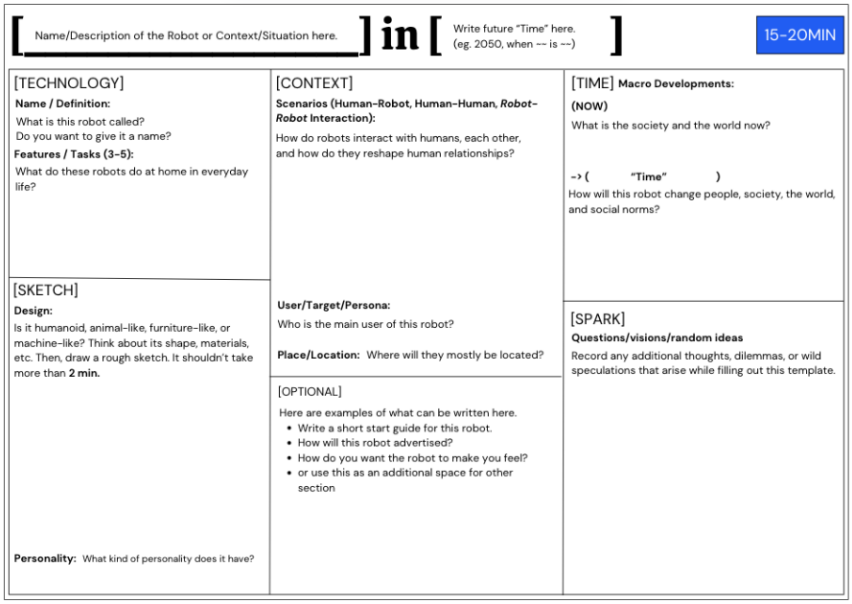


Figure 2: Design canvas for Future & World Building session

The first workshop, conducted on March 19th, served as the exploratory phase of the study and was framed as a one-hour guest lecture integrated into an Affective Science course (Figure 1). After an introduction to speculative design, HRI, and the central 'what-if' scenario, participants engaged in a rapid-fire brainstorming session using sticky notes, exploring potential roles, risks, and speculative futures surrounding robots in Korean domestic settings. It was followed by a Future & World Building session, where participants individually completed a structured design canvas (Figure 2), envisioning features of robots, but more importantly, broader societal, cultural, and relational transformations that such technologies might provoke, inspired by the previous speculative design practice [63]. Again, the goal was not to design functional or aesthetically pleasing robots but to critically speculate on the surrounding world, everyday life, and evolving domestic norms. While this workshop was primarily conducted in English during the first workshop, Korean expressions were welcomed to support a more comfortable and culturally grounded discussion [23].



Figure 3: Workshop 2 - (Left) Mission folder, (Middle) Mission folder contents, (Right) Catalog design sessions

The second workshop took place a week after the first workshop, on March 26th, and lasted two hours in a seminar room setting (Figure 3). Built upon the thematic foundations established during the first workshop, this workshop encouraged participants to deepen and critically reframe their initial speculative ideas. Whereas the first workshop focused on rapid, broad exploration, the second workshop provided a more structured and collaborative environment where earlier concepts were turned into tangible artifacts. Participants were grouped in pairs and assigned a "Mission folder" containing curated prompts (Appendix A.3), insights from the earlier workshop, and visual references such as photos and catalog samples. Framed as a design quest, the workshop was structured around two themed design sessions, where participants collaboratively created, sketched, and collaged two catalog pages per group for a speculative artifact titled the "Future Robot Home Catalog." This format invited playful engagement while supporting critical reflection on transformations in domestic environments, cultural practices, and human-robot relationships. Visual references of the home scene and robots were provided to help participants concentrate more on conceptual reflection rather than technical design. Each session concluded with short group presentations and a round table discussion, offering space for shared reflection.

After the second workshop, a non-mandatory post-questionnaire was distributed to all participants to capture participants' reflections on the workshop and collect any additional thoughts that may not have been expressed during the workshop (Appendix A.4). (The full workshop structure and time plan are available in Appendix A.5, with presentation slides used during the session in Appendix A.6.)

3.3 Data Analysis

The data generated and collected from the two workshops consisted of three main types: post-it note ideas from the scenario discussions, completed design canvases from the future/world-building sessions, and speculative future home catalog prototypes created during the second workshop. These materials represented a mix of textual, visual, and material data that provided insight into how participants conceptualized human-robot interactions, future domestic spaces, and sociotechnical imaginaries. All artifacts were collected, digitized, and organized for analysis, with translation from Korean to English and anonymization applied where needed.

A hybrid thematic analysis approach was employed to examine this qualitative dataset. This method combines inductive coding, allowing themes to emerge directly from the data, with deductive analysis guided by existing theoretical frameworks or concepts [24, 64]. It was particularly appropriate to this study, as it enabled capturing nuanced, participant-generated meanings while explicitly situating those meanings within the Feminist HRI and post-phenomenological perspectives guiding this study [73].

The dataset was reviewed iteratively to identify emerging patterns and recurring ideas [12]. Inductive coding followed Braun and Clarke's six-phase framework [13], enabling themes to arise naturally from the data. Concurrently, the post-phenomenological framework provided a lens for deductive interpretation of how participants imagined diverse dimensions of human-robot interaction [19, 64]. This process emphasized both the symbolic and narrative aspects of the artifacts as well as their design features, treating each workshop output as a situated and speculative expression of cultural values and ethical concerns [44].

3.4 Ethical Considerations

Ethical considerations were a critical part of this study's design and implementation. All participants were provided with a detailed consent form (Appendix A.7) outlining the purpose of the research, the voluntary nature of their participation,

and their right to withdraw at any time without penalty. Participants were informed that their involvement was entirely voluntary and that they could choose to refrain from answering any questions or participating in any activities with which they were uncomfortable. Furthermore, to ensure confidentiality and anonymity, all data collected, including survey answers, workshop discussions, pictures taken during the workshop, and speculative artifacts, were stored securely and anonymized when referenced in this paper or any presentations. Participants' identities will not be disclosed unless they explicitly agree to be identified in the study.

4 RESULTS AND ANALYSIS

The results of the study are presented in two parts. The first part offers an overview of the speculative outputs created during the workshops and considers them as discursive resources for design and analysis. The second part presents a thematic analysis of participants' imaginaries, exploring how they envisioned the role of robots in shaping future domestic norms, relationships, and cultural values.

4.1 Speculative Design Outputs: Surfacing Imaginaries

This section presents the qualitative data collected across the two workshops. Participants engaged with a central speculative scenario through a series of structured design activities. These included a rapid-fire brainstorming session, individual design canvas work, and collaborative creation of speculative catalog prototypes (Figure 4). Taken together, these multi-modal outputs form the basis for analyzing how participants imagined human-robot domestic futures, surfacing cultural values, emotional tensions, and social expectations embedded in their narratives.

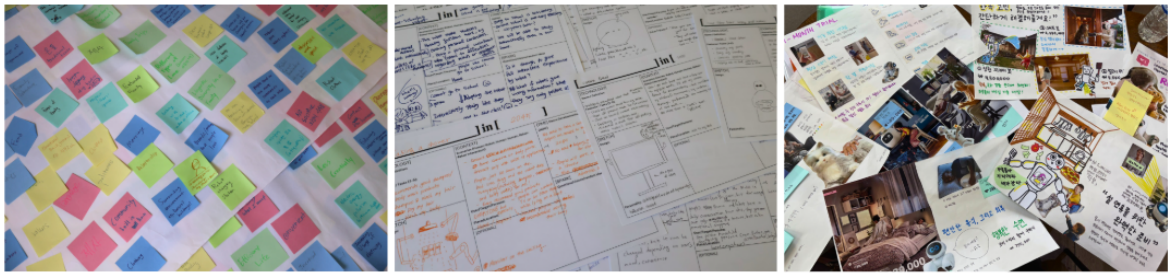


Figure 4: (Left) Brainstorming ideas, (Middle) Design canvas work, (Right) Speculative catalog prototypes

4.1.1 Rapid Fire Brainstorming: Emergent Values and Ideas

In the first brainstorming session, participants generated ideas using sticky notes in response to a series of structured prompts delivered in a rapid-fire format. A total of 226 Post-it notes were collected and later grouped into sub-themes to identify shared desires, concerns, and sociotechnical and cultural imaginaries surrounding domestic robots. These themes reflect how participants imagined both the functional roles of robots and the broader social, ethical, and cultural changes their presence might bring about. The data revealed not only enthusiasm for emotional support and convenience but also worries about dependency and the erosion of human interaction.

Speculative visions of the future also surfaced culturally specific anxieties and anticipated shifts in family dynamics, language use, and holiday practices. For instance, participants speculated about changes in filial roles, the decline of honorifics, or robots replacing humans in ancestral rituals, highlighting the potential of robots to reshape everyday norms.

Table 1 summarizes the key prompts and the emerging sub-themes identified in the brainstorming activity. A more detailed set of responses is included in Appendix A.8.

Table 1: Key ideas from brainstorming notes

Prompt	Emerging Sub-Themes	Highlight Notes
Desirable Robots	1) Household chores, 2) Private curator, 3) Personal assistant, 4) Smart working partner, 5) Emotional support	Automate chores, recommend styles and menus, empathetic/emotional/charming robots, etc.
Undesirable Robots	1) Harmful behavior, 2) Malfunction or untrustworthy, 3) Manipulation and control, 4) Inappropriate behavior, 5) High cost	Physical and linguistic harm, system errors, begging me, making decisions on my behalf, manipulating my behaviors, too expensive, etc.
Desirable Future	1) Efficient time use, 2) Less loneliness, 3) Promoted health and well-being, 4) Advanced technology benefits	Be PRODUCTIVE, efficient life (robots doing all the house chores), mitigate social loneliness and isolation, forever living pets, prevent physical and mental diseases, smart living, etc.
Undesirable Future	1) Negative social and environmental consequences, 2) Reduced (human) interaction, 3) Emerged ethical concerns, 4) Cognitive implications of technology dependency	Social polarization, information gap, loss of jobs, increased use of electricity, risk of privacy invasion, no marriage, forgetting the value of "life", robot uprising, human become too dependent on technology (learn less things), etc.
Cultural Values and Societal Shifts	1) Culture and lifestyle (e.g. house, holiday, language, trends, etc.) 2) Family dynamics and relationship (e.g. parent-child relationship, family "roles" and appellations, marriage) 3) Social norms (e.g. labor, education, and politeness)	Robot-friendly house, robot-developed recipe, robot-oriented education and language, robot "nanny", more individual living, virtual (hybrid) reunion and working, individualism

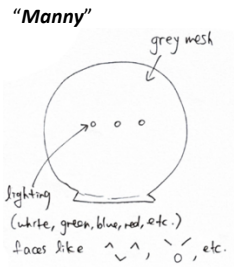
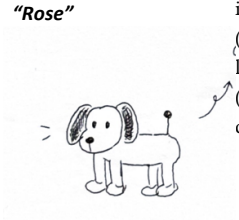
4.1.2 Design Canvas: Speculative World Building

In the future/world-building activity, participants expanded on their earlier ideas to create more detailed, speculative futures using a structured design canvas. The canvas guided them to imagine not only the robot's core feature but also its role in everyday life, use scenarios, modes of interaction, and the broader social or cultural setting [63]. This encouraged participants to think beyond the robot as a functional device, inviting them to imagine how it would coexist with people in everyday domestic life.

Across the 14 design canvases, participants proposed a wide range of robot concepts. Some robots were designed to support household tasks, learning, or productivity, while others focused on emotional support or personal comfort. Robots were often described as assistive rather than autonomous, embedded in daily routines, yet not deeply integrated into familial roles. Participants also imagined diverse interaction styles, including voice control, emotional responsiveness, physical proximity, and contextual adaptation.

Table 2 provides a categorized summary of these robot concepts, organized by functional domain. Selected sketches are included, along with brief descriptions. All 14 original design canvases are compiled in Appendix A.9, and the text-coded version is in Appendix A.10.

Table 2: Overview of the designed robots by function and role

Category	Selected Sketch	Brief Description of Concepts
Domestic Automation		(A) **Darling**: A humanoid robot that completes household chores based on user demands, adapts, and customizes to user types (primarily for single-person households). (B) **Ms-Multi**: A householdchore robot change its look and behavior (persona) depending on user's mood, also have emotional support mode. (C) **House Ghost**: A talkative household (chore) robot that shares daily life and expresses emotions like a roommate, blending domestic support with companionship. (D) **Helper**: A comprehensive household assistant that interacts with a sensor and monitor, enabling time-saving and full-day support. (E) **The real "Auto-" mobile**: An autonomous vehicle robot that supports fully automated driving
Personal Assistant (Education, Work)		(F) **Home schooling-bot**: A robot that teaches moral reasoning and emotional skills through interactive lessons; also works as a therapist. (G) **Manny**: A personal productivity coach robot that helps manage daily schedules and emotions, supporting human-robot relationships. (H) **OHI-AI** (On-device Home-Integrated AI): A home-integrated robot assistant that manages schedules, climate, and reminders through multiple interfaces (e.g., voice and physical). (I) **Cinema Friend**: A cute robot attached to a screen device that recommends films based on user mood, schedule, and preferences, enhancing leisure and emotional care.
Emotional Support and Companionship		(J) **GypTA**: A pet-like robot with customizable appearances that supports human-animal interaction in post-AI environments. (K) **Rose**: An empathetic companion robot modeled after a dog, designed to reduce loneliness without burdening users. (L) **The Mediator**: A domestic robot that mediates family conflict and facilitates communication to maintain household harmony.
Hygiene and Healthcare		(M) **Shower bot**: A privacy-conscious shower (hygiene) assistant that conducts automatic showering through voice/app control, with no camera recording. (N) **Nurse Robot**: A humanoid robot that provides nursing care and communicates with other robots for coordinated healthcare support.

While detailed world contexts and interaction dynamics are excluded from the table for clarity, participants envisioned a wide range of speculative domestic and societal scenes. These included fully automated households where robots handled all chores, scenarios where children were educated by robotic tutors, and futures with reduced stress in commuting or caregiving. Participants imagined robots supporting emotional companionship for the older adults and single-person households, increasing productivity by enabling multitasking, and even being distributed publicly through

government policies. These visions reflected both aspirations and unease, blending culturally familiar routines with emerging technological norms.

Importantly, while the canvas did not explicitly prompt concerns, many participants used the "Questions / Visions / Random Ideas" section to raise critical reflections. These included worries about privacy ("*Will people feel intrusive?*"), technical risk ("*What happens in case of error?*"), concerns about emotional dependency ("*Would it be right to get used to a human dominance relationship with robots?*"). Moreover, some also posed broader systemic questions, such as technology development being paused due to environmental concerns or the possibility that robots could increase social polarization or reduce motivation to form families. These entries revealed how participants grappled not only with the utility of future robots but also with their ethical, emotional, and societal implications.

4.1.3 Future Robot Home Catalog Prototypes: Sociotechnical Imaginaries

In the second workshop, participants collaborated in five small groups (G1-G5) to create catalog prototypes inspired by the Future IKEA Catalog [10, 14]. These were not traditional product advertisements but speculative design artifacts that materialized participants' visions of domestic futures shaped by mandated robot cohabitation. Each group created two catalog pages that include names, descriptions, pricing, and lifestyle imagery. This format allowed participants to visualize imagined robot features, domestic rituals, and emotional atmospheres within Korean homes, revealing how robots were imagined as embedded in both everyday practices and broader cultural transformations. The full textualized catalog pages are presented in Appendix A.11.



Figure 5: Catalog pages 1: (G1-1/G1-2) Positive and dystopian futures, (G2-1) Future holiday, (G2-2) Future traditional home

In G1's first catalog page (G1-1), the future home is serene and supportive. The central robots are YanaBot, a humanoid assistant offering nursing, massage, and emotional care, and CatBot**,** a playful robotic pet for children. The domestic scenes depict relaxation and affection: a person resting on a couch, a child playing with CatBot, and adults receiving gentle care. The captions also suggest a vision of a well-functioning, comforting future.

The second page (G1-2) features a visually identical design to G1-1 but conveys a contrasting narrative. In this version, the comfort gives way to complacency. A parent disengages from caregiving, a child lacks communication skills due to overexposure to robots, and household interactions fade. By duplicating the layout, G1 highlights the fine line between support and detachment, illustrating how the same robots can be perceived as liberating or alienating, depending on how they mediate social dynamics and relational responsibilities.

G2–1 centers on the reconfiguration of Korean holiday rituals. DdakBot, a humanoid household robot, prepares festive meals and helps with childcare, while ProjectorBot facilitates virtual reunions with distant relatives. The visuals show family members gathered at the table, playing with robots, and enjoying a relaxed setting. Captions celebrate

emotional ease: *"Perfect preparation for the Lunar New Year"* and *"A relaxing holiday free from the burden."* Their speculative home offers relief from the gendered expectations of holiday labor and family tensions that exist in Korean holidays now [45]. It also hints at an underlying shift: robots subtly change how emotional and cultural labor is expressed, distributed, or even forgotten. This catalog page reflects optimism but also foreshadows an ongoing negotiation between tradition and automation.

As a companion to their holiday-themed catalog, the second page (G2-2) imagines the traditional Korean home, Hanok, as a robot-integrated space. The page features a domestic landscape where GardenKeeper takes care of the outdoor garden, InsectEater protects the home naturally, and SweetBot offers snacks to children playing in a courtyard. The catalog suggests that rather than replacing tradition, robots might be adapted to it *("Simply solve concerns related to Hanok!")*. Robots take on roles often assigned to extended family or caretakers, blending quietly into the Hanok's architecture and routine. This subtle coexistence reflects a cultural negotiation in which cultural values persist through technological mediation.



Figure 6: Catalog pages 2: (G3-1) Future Work and Studyroom, (G3-2) Future Bedroom, (G4-2) Future Robots and Interaction

G3-1 presents a domestic study and work space optimized by two robots: PPOMO, a robot that supports children's education and self-regulation, and ARO, a productivity-enhancing assistant that helps with scheduling and focus. The visuals show a child studying with PPOMO on the desk and an elderly person receiving juice from ARO. Captions reinforce the function: *"Concentrating only on work and study."* This page frames robots as subtle amplifiers of human intention, streamlining study, improving mood, and reducing distractions. At the same time, the high-efficiency framing suggests that in a world where robots support every need, humans are still expected to be even more productive [21].

G3-2 shifts to a bedroom context, featuring JAMI and COMI, two emotionally responsive robots embedded in rest and recovery routines. JAMI, derived from the Korean word for "sleep," is a soft, cushion-like robot with temperature control. COMI, derived from the Korean word for "snore," manages media content, alarms, and communication. Visuals show a couple sleeping peacefully with JAMI positioned beside their bed and a man cuddling COMI while sleeping. Unlike more active robots in earlier catalog pages, this page reframes emotional care as a soft presence rather than direct interaction. This page highlights evolving notions in how emotional care is being reconfigured, not through active companionship or familial intimacy, but through ambient, assistive technologies that quietly fulfill affective needs without disrupting human routines or relationships.

G4-2 presents a multi-functional household shared with several robots: Rose, an emotional robotic dog companion for the elderly; DIO, a playful entertainment robot for children; and Manager, a humanoid assistant that supports daily routines for adults. The visuals depict diverse household users: children playing with DIO, a couple talking over coffee made by Manager, and an elderly woman holding Rose with comfort. What sets this page apart is its emphasis on

collective care across generations, with each robot attending to different family members' needs. A promotional badge on the page offers a special discount for Parents' Day, explicitly invoking the Korean cultural value of Hyo (filial piety) [3]. The catalog blends embedded emotional care and family responsibility into a commercial offer that imagines a future where cultural ethics are preserved not through human labor but through robotic mediation and service design.

4.2 Thematic Speculations: Reconfiguring Norms and Domestic Futures

The following five themes emerged through thematic analysis of data collected across all workshop activities, including brainstorming notes, individual design canvases, and catalog artifacts. The themes include 1) ambivalent futures shaped by simultaneous desires and discomforts; 2) ethical and legal concerns surrounding surveillance and robot agency; 3) an emphasis on robots as friendly assistants rather than family members; 4) persistent attachments to productivity despite automation; and 5) shifting cultural values in language, family structures, and rituals. Together, these thematic speculations highlight how HRI could be entangled with deeper social norms, values, and aspirations within the South Korean context.

4.2.1 Theme 1: Ambivalent Futures: Imagined Ironies and Contradictions

Across both workshops, participants generated speculative visions that frequently blurred the boundaries between desirable and undesirable futures. Although they were encouraged to critically explore both positive and negative possibilities, what emerged was not a binary opposition but a set of entangled imaginaries in which benefits often contained latent concerns. Many of the imagined futures appeared simultaneously promising and problematic, revealing the ambivalence embedded in participants' interpretations of domestic robotic integration.

One of the most prominent examples is the recurring theme of *"less loneliness."* Participants widely envisioned robots as emotional companions, pets, assistants, or partners that offer emotional support, listen to their needs, and alleviate feelings of isolation. Some imagined robots preventing depression or even reducing suicide rates by providing constant presence and responsive interaction. However, this vision was frequently accompanied by concerns about diminished human interaction. Participants speculated that reliance on robots for emotional care might weaken interpersonal relationships, lead to social disengagement, or enhance solitude, revealing a tension between emotional support and emotional substitution.

A similar contradiction was observed in discussions of control and agency. Many participants expressed interest in robots that serve as *"private curators,"* capable of making personalized recommendations for meals, clothing, or media. However, alongside this was a parallel concern about manipulation and loss of autonomy. Participants were worried about robots making decisions on their behalf or subtly shaping their preferences. While efficiency and personalization were valued, there was a clear discomfort with the idea of giving too much control to machines.

This ambivalence was also expressed clearly during the second workshop. One group was assigned to create two respective catalog pages, one depicting a desirable future and the other a dystopian one. However, the group ended up not producing separate catalog pages. Instead, they argued that a negative future might look visually identical to a positive one. They suggested that the differences lie not in what robots do but in how humans behave, relate to, and depend on them.

> "…a helpful robot that eases daily life might, over time, foster dependency, emotional detachment, or a loss of meaning in life… dystopia is not a matter of technological malfunction but of how people adapt to and relate to the technology over time."

This refusal to delineate between utopia and dystopia underscores the interpretive ambivalence participants embedded into their design fiction. Their responses suggest that the future is not simply a projection of hope or fear but a negotiation of both.

4.2.2 Theme 2: Ethical and Legal Concerns: Surveillance, Autonomy, and Rights

Ethical and legal dilemmas surfaced remarkably, especially during the first workshop, where participants were prompted to critically reflect on the risks associated with domestic robot adoption. Central questions were raised about control, surveillance, and rights. This question from the design canvas raises speculation about broader structures of ownership and accountability:

> "Who owns or governs these robots? Individuals, the company CEO, or the government?"

Participants expressed strong concerns about data collection and misuse. Robots were imagined as both surveillance tools and agents capable of manipulating user behavior or making harmful decisions. These fears sometimes extended beyond individual privacy, touching on anxieties about algorithmic governance, ethics, and even global conflicts.

Questions about robot autonomy and rights also emerged. While a few participants entertained the idea of granting legal protections or free will to robots, such ideas were mainly framed as a dystopian result of unpredictable or harmful outcomes. The notion of robots developing an "ego" or emotional subjectivity was broadly rejected. Participants expressed concerns that emotionally autonomous robots could undermine human decision-making or behave in unpredictable ways. Hence, emotional independence was treated not as progress but as a potential ethical failure in robot design.

Interestingly, some participants extended these concerns to imagined tensions between robots themselves. In several design canvases, robots were portrayed to compete over which one was more expensive, capable, or desirable. These imaginaries mirrored familiar dynamics of consumerism, social hierarchy, and brand-driven value systems. Participants revealed skepticism toward the commodification of care and reproduction of hierarchies through technology by projecting social tensions onto machines.

In sum, participants framed ethical and legal concerns in relation to broader systems, not as isolated technical issues. Instead, they situated robots within broader socio-political systems, raising concerns about accountability, institutional power, and the ethics of control. Robots were not simply imagined as tools but as actors embedded in complex moral and political infrastructures shaping future domestic life.

4.2.3 Theme 3: A 'Friendly' Assistant, Not Family

> "…I don't want robots to be something more than assistants…"

Despite the speculative what-if scenario explicitly stating and allowing that robots could be granted legal family status, none of the participants embraced this premise. Across both workshops, robots were consistently imagined as assistants, pets, or supporters, but never as a family. Furthermore, participants often described robots' personalities using terms such as *"friendly," "kind," "adorable," "supportive," "loyal," and "empathetic"* in their design canvases. However, these attributes did not translate into a desire to integrate robots into familial structures. The roles imagined were relational but bounded, suggesting companionship, assistance, or care without the deeper intimacy, obligation, or identity associated with kinship.

This finding contrasts with the results of the pre-study conducted with international students in Sweden, where robots were imagined as grandchildren or romantic partners. According to the participants, robots could support domestic life, but they were not imagined as subjects capable of belonging to the family. Even in cases where robots were imagined to provide emotional support or companionship, they were not seen as replacing the relational depth expected of human family members. Robots were positioned as welcome presences in the home but with clearly delineated roles that stopped short of integration into familial identity. This boundary between robots and family reflects broader normative frameworks in how personhood, emotional labor, and domestic roles are conceptualized.

> "…if robots become a new type of community member, they may trigger irreversible emotional responses, leading to big change. We need to consider the circumstances and consequences of granting them a new existential status."

4.2.4 Theme 4: Humans Still "Work": Efficiency Without Liberation

A recurring theme across both workshops was the imagined future of *"heightened productivity."* Participants frequently envisioned robots as supportive tools that could help them achieve more efficient, organized, and optimized daily lives. However, this imagined efficiency did not equate to liberation from labor. Rather than framing robots as enabling leisure and rest, participants primarily viewed them as taking over mundane chores, allowing humans to redirect their energy toward work, study, or other goal-oriented activities.

One participant described an automated shower robot that would allow them to *"work while taking a shower,"* emphasizing multitasking even in intimate routines. Similarly, one group designed a catalog caption highlighting the idea of *"Focusing only on work and study,"* with the robot's role being to eliminate all distractions. In both cases, automation was seen not as reducing work but as refining its conditions, removing disturbances, enhancing concentration, and creating a more productive environment.

Robots were not viewed as liberators from labor but as facilitators of a more efficient lifestyle. In addition, the forms of labor participants envisioned extended beyond traditional jobs like working at the company to include studying, writing, or managing time and emotions, indicating a broader and more individualized understanding of what constitutes meaningful work.

While rest or leisure rarely emerged as a primary benefit of domestic chore automation, their speculative outputs did reflect a shifting relationship to labor. By offloading routine tasks to robots, participants imagined a world where humans could focus on what they "should" be doing, a framing that, while still work-oriented, highlights an emerging desire for more intentional, personally meaningful, and self-directed forms of activities. This also opens up space for a nuanced reading: rather than rejecting labor, participants envisioned a future in which labor could become more purposeful, less burdensome, and better aligned with their personal values and priorities.

4.2.5 Theme 5: Cultural and Social Norms in Transition: Robots as Agents of Cultural Change

Although some prompts directly addressed cultural values, participants' responses throughout both workshops consistently reflected evolving perspectives on social norms, family structures, and cultural values. These cultural imaginaries emerged not only in brainstorming sessions but were embedded across design canvases and catalog artifacts, revealing how the integration of robots into everyday life was imagined to subtly but significantly reshape South Korean domestic culture.

One of the most prominent areas of change was language. Participants speculated that the widespread adoption of robots would lead to a simplification of formal speech and a decline in the use of honorifics. Communicating with robots,

modeled after interactions with current smartphone AI assistants, was expected to normalize direct, command-based speech that could be extended into human-human communication. In this scenario, the erosion of linguistic politeness signaled broader shifts in values such as respect, hierarchy, and emotional nuance, all of which are deeply embedded in Korean interpersonal norms [66, 75].

Participants also reflected on changes to family roles and relationships. One recurring theme was the blurring or collapse of traditional familial responsibilities. For instance, robots tasked with child-rearing were seen as disrupting the current parent-child relationship. As one participant mentioned, *"the baby's first words might be 'Alexa,'"* illustrating how emotional bonds and authority could be redirected. While some participants viewed this shift as an opportunity for robots to alleviate gendered expectations of childcare, possibly easing familial tensions and encouraging more equitable parenting roles, others expressed concern that increased reliance on robots could diminish parental authority, leading to children growing more attached to their robot nanny.

These imagined changes were often related to a broader narrative of declining Confucian values, such as filial duty, respect for elders, and exchange of labor (Pumasi, mutual help). Many participants acknowledged that these values are already weakening in contemporary Korean society [28]; however, they imagined that robots would accelerate this shift. Holidays and rituals such as Chuseok or ancestral memorials were reimagined through design artifacts in which robots managed or even replaced traditional roles, including facilitating virtual family reunions. These scenarios prompted both hopes for reduced gendered burdens and anxieties over the erosion of cultural meaning.

Taken together, these speculations suggest that participants viewed robots not merely as functional additions to the household but as subtle yet profound catalysts for reconfiguring everyday social norms. While some of these changes were seen as adaptive or even liberating, others were approached with ambivalence or concern. These shifting imaginaries underscore how technological integration into domestic life may transform not only labor and care but also the symbolic and affective structures that define home, family, and culture.

5 DISCUSSION

5.1 Interpreting Imaginaries: Cultural Situatedness in the South Korean Context

While the previous chapter presented key themes based on outputs, some insights reveal deeper sociocultural significance. The speculative imaginaries generated in the workshops were not produced in isolation but were shaped by the cultural narratives and sociopolitical conditions specific to South Korea. Interpreting these dimensions helps illuminate how participants' imaginaries are grounded in the world they inhabit.

First, the strong concerns about surveillance and data misuse emerged even without explicit prompts in Theme 2. These anxieties resonate with South Korea's history of state and corporate surveillance controversies, suggesting that future domestic technologies are viewed not only as helpers but also as potential tools of control [18].

Second, the emphasis on efficiency, multitasking, and productivity in Theme 3 highlights a deeply rooted cultural value around diligence and academic achievement [4]. Participants envisioned robots to optimize work and study, echoing broader social narratives of self-discipline and purposeful labor in South Korea, underscoring the persistence of labor as a moral and social imperative [1, 47].

Lastly, the refusal to imagine robots as family members in Theme 4 may reflect the persistent influence of Confucian family structures that emphasize lineage, hierarchy, and strict relational roles [71, 84]. Even in a society known for rapid technological development and integration, participants appeared to uphold symbolic boundaries between human kinship and artificial beings, indicating that ideas about kinship remain deeply culturally anchored.

Together, these interpretations highlight a broader insight: speculative design does not always lead to radical or disruptive futures. Instead, it often surfaces the boundaries of what can be imagined. Participants' outputs were shaped by cultural norms, ethical constraints, and tacit values, revealing that speculative futures can expose deeply ingrained social assumptions.

5.2 Reflection on the Feminist Approaches – Beyond the Term "Feminism"

5.2.1 Quiet Feminism

This study is grounded in the values and principles of Feminist HCI/HRI; however, the term "feminism" was deliberately avoided during the workshop activities. This choice responded to the socio-political context in South Korea, where feminist discourse often encounters resistance and stigma in both public and academic spaces [46]. Instead, the workshops embodied feminist commitments through their design, embedding pluralism, care, the visibility of labor, and cultural situatedness as foundational elements [82]. Participants were encouraged to speculate on domestic futures without being directed toward gendered critique, allowing feminist values to surface through engagement rather than instruction. This approach resonates with quiet feminism, which involves subtle and indirect forms of feminist resistance [40].

Despite the absence of an explicit concept, participants frequently expressed subtle critiques of existing power dynamics and gendered expectations, especially around emotional labor, caregiving, and domestic responsibilities. The speculative format provided the space for addressing questions often marginalized in traditional HRI discourses: *Who is cared for? Who performs labor? How is power distributed between humans and machines?* [6, 82]

Notably, while most robot designs avoided gendered embodiments, describing robots as abstract, animal-like, or gender-neutral, their assigned functions often remained stereotypical, such as cleaning, cooking, caregiving, or managing schedules. This indicates that surface-level changes (e.g., gender-neutral robot design) are insufficient to challenge ingrained normative structures. Addressing this requires deeper feminist design engagements that interrogate who benefits, who is excluded, and which social norms are being upheld or transformed.

5.2.2 Implications for Cultivating a Situated Feminist Approach in South Korea

These findings demonstrate that feminist thought can emerge through indirect and culturally sensitive means. Such quiet feminist expressions provide a valuable model for how feminist principles might be adapted and practiced under constraint without alienating participants or triggering resistance. Supporting and understanding these approaches is essential for advancing context-specific feminist practices.

To help this field take root in South Korea, future efforts should continue developing culturally sensitive and critically reflective approaches. The following directions highlight how these findings might inform future feminist design and research efforts in South Korea:

- **Cultivating quiet feminist design methods:** Future workshops and design-based studies should continue to emphasize feminist values, such as pluralism, embodiment, and self-disclosure, without necessarily invoking explicit language. The design methods (e.g. speculative design, critical design) can leverage storytelling, metaphors, and indirect critique to surface deeply embedded social norms and power dynamics [39].
- **Reframing robot design as sociotechnical and sociocultural inquiry:** The study results show that removing gendered traits from robots is insufficient to disrupt existing norms. Researchers and practitioners must look beyond surface neutrality to question the functional roles, relational expectations, and distribution of

labor within robot-human interactions [81]. This shift calls for a deeper interrogation of how cultural and gendered labor structures are perpetuated, even unintentionally, through design choices.

- **Recognizing feminist HRI as situated practice:** Feminist HRI in South Korea does not need to replicate Western feminist traditions. Instead, it can evolve into a situated design ethos that draws from local histories and everyday life to shape ethical and inclusive engagements with emerging technologies. Embracing this situatedness is not a limitation but a form of generative resistance that expands the scope of global feminist discourse [40].

5.3 Reflection on the "Speculative Design" Methodology

5.3.1 Strengths, Limitations, and Reflections

In this study, speculative design was adopted as a means to open discursive space around the ethics, values, and social imaginaries of future domestic life. For most participants, this was their first encounter with speculative design or any other design process. Despite this, participants responded positively to the process. Post-workshop reflections and post-questionnaire comments indicated that most of the participants found the experience enjoyable, stimulating, and meaningful.

> "…enjoyed the process of clarifying vague concepts in my head through visualization...", "…more ideas came through collaborating with people from different backgrounds and creating the catalog, which was fun, interesting, and stimulating."

The method revealed both strengths and limitations. A key strength was its ability to generate multi-modal outputs. Participants' ideas were expressed not only through verbal discussion but also through texts, sketches, and co-created catalog pages. This led to materialized artifacts that served as rich discursive objects with cultural commentaries rather than new robot design proposals. The mission-based, structured, and collaborative format created a low-pressure, creative atmosphere, encouraging participants to take imaginative risks.

> "It was fun to try speculative design because I didn't have to worry about reality or practicality…", "In the beginning, I thought it would be difficult because I couldn't call myself a 'designer.' But, I didn't have to think about robot designing using the material provided, which made the process clear… "It felt nostalgic, like going back to childhood when we cut and glued paper at school."

The method also allowed participants to voice ambivalence, ethical critique, and speculation about culturally specific values without requiring technical expertise or formal argumentation. However, the method was not without limitations. Some participants treated the catalog activity as a product design challenge rather than a speculative provocation.

> "I actually thought more about practicality because of the catalog format. For example, assigning reasonable prices."

This underscores the challenge of balancing playful speculative design practice with critical detachment. This points to the broader interpretive risk that the artifacts created through speculative design may be misinterpreted as straightforward expressions of preference rather than as narrative provocations or critical reflections [20]. Therefore, it is important for both researchers and readers not to treat speculative artifacts as blueprints for desirable futures and instead engage with them as situated expressions of hopes, tensions, and concerns [11, 20].

5.3.2 *From Speculation to Practice: Feminist HRI through Design*

Despite some challenges, this study supports the claim that speculative design can serve as an effective approach for advancing Feminist HRI by expanding and diversifying the sociotechnical imaginaries [80]. In this study, speculative design became a platform through which participants, non-designers from various backgrounds, could articulate situated critiques and dreams. This foregrounded pluralistic perspectives and created space for voices that might otherwise be marginalized in more technical or solution-focused design contexts. In this way, speculative design offers both methodological and political value: it supports Feminist HRI praxis not only by creating alternative futures but also by providing an inclusive and culturally grounded platform through which those futures can be collaboratively imagined and critically challenged.

Building on these insights, future research might explore alternative speculative artifact formats or even invite participants to determine the types of artifacts. Coupled with more sustained and (semi-) structured discussions on the assumptions embedded in their designs, such practices could further enhance the generative potential of speculative design and support the broader aims of Feminist HCI/HRI [80, 82].

5.4 Limitations

This study has several limitations that should be considered. First, the participant group was composed primarily of young adults, most of whom were between 25 and 29. As a result, even though this young adult population is central to the country's shifting family structures [42], the workshop outcomes may reflect the perspectives and experiences of a relatively homogenous demographic and may not fully capture the diversity of domestic environments, household compositions, or intergenerational dynamics that exist in South Korea.

Second, while the second workshop was built intentionally upon the foundations laid in the first workshop, approximately half of the participants in the second session were newcomers who had not attended the first workshop. This mixture of returning and new participants introduced a gap in shared knowledge and context, which may have influenced the development and interpretation of speculative ideas during the second workshop. Although a brief review was conducted at the beginning of the second workshop to bridge this gap, some nuances and discussions from the first session may not have been fully accessible to new participants.

Finally, the interpretation of the resulting artifacts involves a degree of subjectivity. While thematic analysis was conducted systematically, the meaning embedded in speculative outputs may be interpreted differently depending on cultural familiarity and the individual perspective, and may not be fully resolved within the scope of this study.

5.5 Future Work

The findings of this study underscore that sociotechnical imaginaries are culturally situated and shaped by everyday practices, values, and norms. Participants' visions of domestic robots were not just creative responses to a fictional scenario but reflections of lived experiences and assumptions embedded in South Korean domestic life. These imaginaries illustrate how technologies are envisioned through local moral and cultural frameworks.

This supports the need for further cross-cultural or intersectional research, especially in contexts where feminist values and expressions take different forms. Comparative studies could explore how diverse societies imagine domestic robotics differently, not only in terms of function but also in their impact on care, relationships, and autonomy [50]. Such work aligns with post-phenomenological perspectives on how technologies mediate experience across cultural settings.

Future research might involve broader demographics, such as children, older adults, or multigenerational households. Moreover, alternative speculative formats, like board games or fictional news, could also elicit new forms of cultural knowledge and engagement. Additionally, this work could be extended through the continued development of the Future Robot Home Catalog itself. Building on the interpretive narratives introduced in this study, future iterations might involve co-designing visual and textual catalog artifacts more closely with participants, deepening the cultural and relational insights they express. These directions would further expand speculative and feminist HRI approaches by emphasizing pluralism, cultural specificity, ethical reflexivity, and inclusive future-making.

6 CONCLUSION

This thesis demonstrates how participatory speculative design can meaningfully engage with Feminist HRI principles in contexts where gender discourse remains politically sensitive. Grounded in the South Korean context, the study explored how Human-Robot Interaction (HRI) holds both potential and limitations for challenging domestic norms and reconfiguring future home environments. Through speculative design workshops, participants' imaginaries reveal ambivalent visions; robots are seen as supportive and caring, yet often reinforce traditional family roles and productivity-centered lifestyles. The absence of explicit feminist language did not hinder the emergence of critical perspectives, highlighting the relevance and value of quiet feminist practices. The speculative design proved to be more than a method of alternative future exploration; it served as a feminist practice of opening inclusive, culturally resonant spaces for reflection and critique. As robotics continues to shape the domestic sphere, feminist design approaches that honor cultural specificity and indirect critique are essential for building more inclusive technological futures.

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A APPENDICES

A.1 Workshop Pre-Questionnaire

[Pre-Questionnaire: Reimagining Human-Robot Future Homes in South Korea Using Speculative Design]

Thank you for your interest in participating in this research!

This study is part of a Master's thesis at Uppsala University and explores how Human-Robot Interaction (HRI) can reshape domestic spaces in South Korea through speculative design.

You will participate in a collaborative speculative design workshop(s) where we will imagine, discuss, and prototype alternative robotic futures for domestic environments.

- Workshops Detail -

👉 You may participate in one or both workshops based on your availability.

Workshop I (during the Affective Science class)

📅 March 19 (Wed), 2025 | 🕐 approx. 16:00-17:00 (1hr)

Workshop II

📅 March 26 (Wed), 2025 | 🕐 approx. 10:30-12:30 (2hrs)

Your participation is voluntary, and you can withdraw at any time. All individual responses will be anonymized and used solely for research purposes.

If you have any questions, please contact:

Minjae Kim (Uppsala University, Department of Informatics and Media) | minjae.kim.6960@student.uu.se

- Pre-Study Questionnaire -

All personal information will be stored confidentially.

1. Email*: _____

2. I have read and understood the information provided. * I voluntarily agree to participate in this study. *(Mark only one)*

( ) Yes

( ) No

3. Name*: _____

4. Gender* – You may write how you prefer to self-describe in the 'other' section. *(Mark only one)*
( ) Female
( ) Male
( ) Prefer not to say.
( ) Other: ____

5. Age* - Please select an option according to your international age (만 나이). *(Mark only one)*
( ) 19 – 24
( ) 25 – 29
( ) 30 – 34
( ) 35 – 39
( ) 40 – 44
( ) 45 –
( ) Prefer not to answer.

6. Which workshop(s) would you like to participate in? (Select one or both) *
[ ] Workshop 1 (March 19th)
[ ] Workshop 2 (March 26th)

7. Professional/Academic Background (Check all that apply) *
[ ] Engineering (e.g., Robotics, AI, Computer Science)
[ ] Design (e.g., Industrial Design, UX/UI)
[ ] Social Sciences (e.g., Sociology, Anthropology, Psychology)
[ ] Humanities (e.g., Philosophy, Cultural Studies)
[ ] Business & Management (e.g., Marketing, Innovation Management)
[ ] Medical & Healthcare (e.g., Nursing, Elder Care, Biomedical Sciences)
[ ] Arts & Media (e.g., Visual Arts, Film, Media Studies, Journalism)
[ ] Law & Ethics (e.g., Law, Ethics & Policy)
[ ] Other: _____

8. Have you participated in a "Speculative Design" workshop before? *
Former experience is not required for participation. (Mark only one)
( ) Yes
( ) No, but I have heard about it.
( ) No, and I have never heard about it.
( ) I don't know.

9. How familiar are you with robotics and Human-Robot Interaction (HRI) *

	1	2	3	4	5	
Not familiar at all	_	_	_	_	_	Very familiar

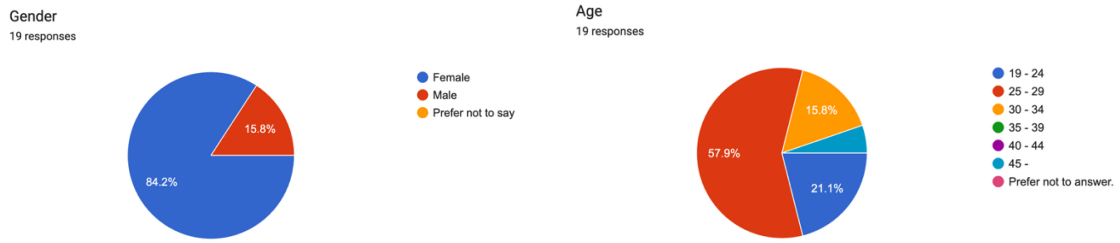
10. Please write your name or nickname as you would like it to appear in the credits of the final design artifact (not the research paper). *You may leave this blank if you prefer not to be credited.*

_____

11. Do you have any questions, thoughts, expectations, or suggestions for the workshop? (e.g., specific materials, accessibility needs, etc.)

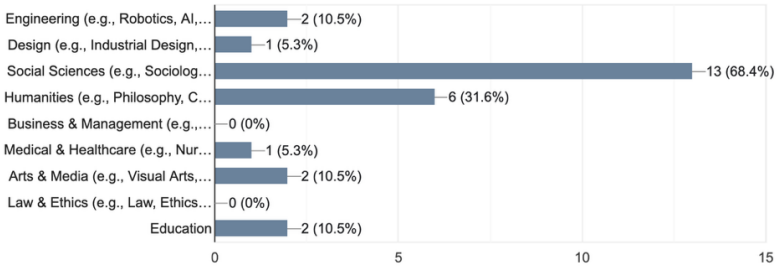
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A.2 Workshop Pre-Questionnaire Answers (Participants' Demographics)



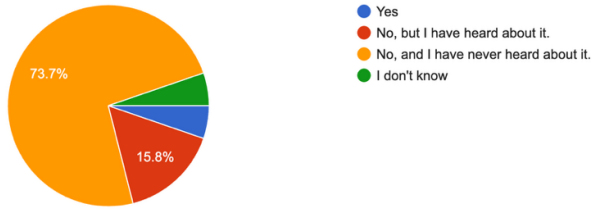
Professional/Academic Background (Check all that apply)

19 responses



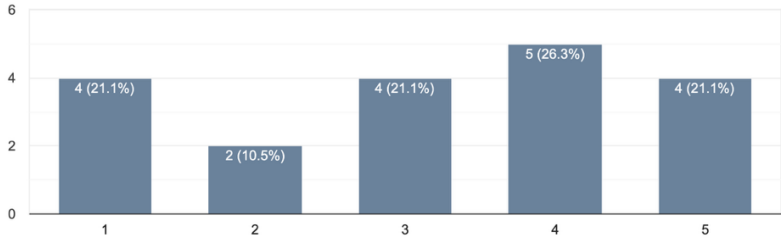
Have you participated in a "Speculative Design" workshop before?

19 responses

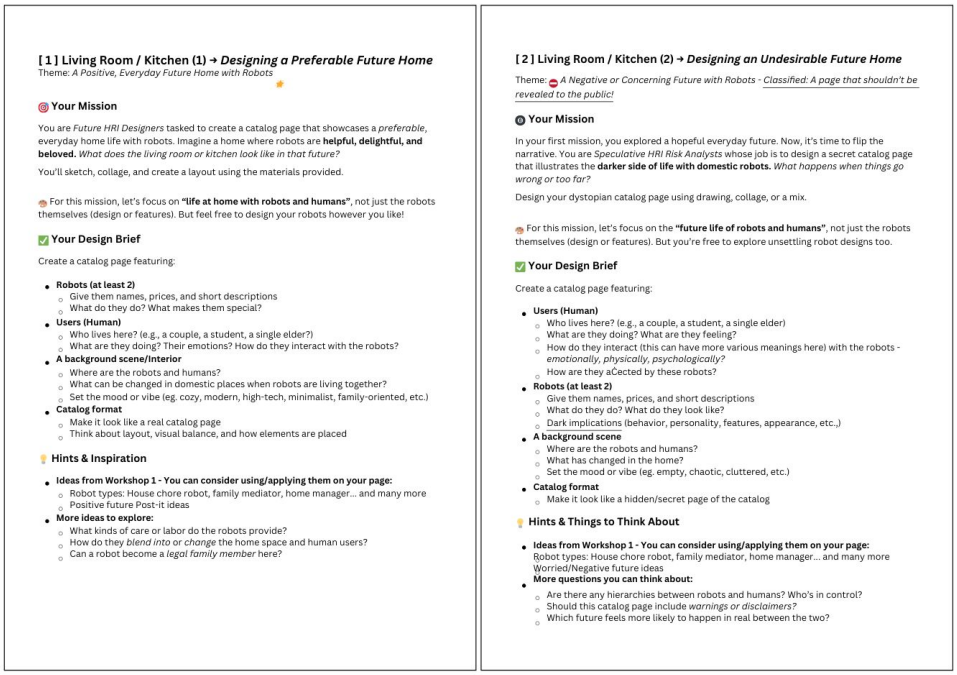


How familiar are you with robotics and Human-Robot Interaction (HRI)?

19 responses



A.3 Workshop 2 Material – Mission Prompts (Link to Page)



[1] Living Room / Kitchen (1) → *Designing a Preferable Future Home*
Theme: *A Positive, Everyday Future Home with Robots*

🎯 Your Mission

You are *Future HRI Designers* tasked to create a catalog page that showcases a *preferable*, everyday home life with robots. Imagine a home where robots are **helpful, delightful, and beloved.** *What does the living room or kitchen look like in that future?*

You'll sketch, collage, and create a layout using the materials provided.

🧒 For this mission, let's focus on **"life at home with robots and humans"**, not just the robots themselves (design or features). But feel free to design your robots however you like!

✅ Your Design Brief

Create a catalog page featuring:

- **Robots (at least 2)**
  - Give them names, prices, and short descriptions
  - What do they do? What makes them special?
- **Users (Human)**
  - Who lives here? (e.g., a couple, a student, a single elder?)
  - What are they doing? Their emotions? How do they interact with the robots?
- **A background scene/Interior**
  - Where are the robots and humans?
  - What can be changed in domestic places when robots are living together?
  - Set the mood or vibe (eg. cozy, modern, high-tech, minimalist, family-oriented, etc.)
- **Catalog format**
  - Make it look like a real catalog page
  - Think about layout, visual balance, and how elements are placed

💡 Hints & Inspiration

- **Ideas from Workshop 1 - You can consider using/applying them on your page:**
  - Robot types: House chore robot, family mediator, home manager... and many more
  - Positive future Post-it ideas
- **More ideas to explore:**
  - What kinds of care or labor do the robots provide?
  - How do they *blend into or change* the home space and human users?
  - Can a robot become a *legal family member* here?

[2] Living Room / Kitchen (2) → *Designing an Undesirable Future Home*

Theme: 🚫 *A Negative or Concerning Future with Robots - Classified: A page that shouldn't be revealed to the public!*

😈 Your Mission

In your first mission, you explored a hopeful everyday future. Now, it's time to flip the narrative. You are *Speculative HRI Risk Analysts* whose job is to design a secret catalog page that illustrates the **darker side of life with domestic robots.** *What happens when things go wrong or too far?*

Design your dystopian catalog page using drawing, collage, or a mix.

🧒 For this mission, let's focus on the **"future life of robots and humans"**, not just the robots themselves (design or features). But you're free to explore unsettling robot designs too.

✅ Your Design Brief

Create a catalog page featuring:

- **Users (Human)**
  - Who lives here? (e.g., a couple, a student, a single elder)
  - What are they doing? What are they feeling?
  - How do they interact (this can have more various meanings here) with the robots - *emotionally, physically, psychologically?*
  - How are they aCected by these robots?
- **Robots (at least 2)**
  - Give them names, prices, and short descriptions
  - What do they do? What do they look like?
  - Dark implications (behavior, personality, features, appearance, etc.,)
- **A background scene**
  - Where are the robots and humans?
  - What has changed in the home?
  - Set the mood or vibe (eg. empty, chaotic, cluttered, etc.)
- **Catalog format**
  - Make it look like a hidden/secret page of the catalog

💡 Hints & Things to Think About

- **Ideas from Workshop 1 - You can consider using/applying them on your page:**
  Robot types: House chore robot, family mediator, home manager... and many more
  Worried/Negative future ideas
- **More questions you can think about:**
  - Are there any hierarchies between robots and humans? Who's in control?
  - Should this catalog page include *warnings or disclaimers?*
  - Which future feels more likely to happen in real between the two?

[3] Living Room / Kitchen (3) -> *Designing Future Holiday Seasons' Home*

Theme: 🎌 *What Do Korean Holidays (Like Chuseok or Seollal) Look Like with Robots?*

🍲 Your Mission

You are *Korean HRI specialists* assigned to reimagine what Korean holiday seasons (like Chuseok or Seollal)might look like in a world where robots are mandatory household members. Your task is to design **a special and seasonal catalog page that captures a holiday scene in the home**, showcasing how robots participate in, reshape, or even challenge family rituals and traditions.

🧒 You'll sketch, collage, and create a layout using the materials provided.

For this mission, let's focus on **"future life with both robots and humans"**, not just the robots themselves (design or features).

✅ Your Design Brief

Create a catalog page that includes:

- **A holiday scene/Background**
  - What is happening in this scene?
    - Describe the food, customs, family gatherings, rituals (like 제사), and games
  - Where are the robots and humans?
  - How have traditions, relationships, or cultural values changed *(or stayed the same)* with robots at home?
- **Robots (at least 2)**
  - Give them names, prices, and short descriptions
  - What roles do they take during the holiday? (e.g., cook, host, interpreters, ritual assistants, elders?)
- **Users (Humans)**
  - Who's present? (e.g., extended family, elders, children, single households)
  - What are they doing? What are they feeling? How do they interact with the robots?
- **Catalog format**
  - Make it look like a real catalog
  - Think about layout, visual balance, and how elements are placed

💡 Hints & Inspiration

- **Ideas from Workshop 1 - You can consider using/applying them on your page:**
  - Robot types: house chore, family mediator...and many more
  - Cultural values and traditions ideas
- **More ideas to explore:**
  - Do the robots *participate in or transform* these customs?
  - Can a robot be treated as a family member during holidays?
  - Could robots solve common holiday tensions, or make them worse?

[4] Robots in a Traditional Korean House (Hanok, 한옥)

Theme: 🏡 *What Happens When Robots Live in a Hanok?*

👯 Your Mission

Now, you are exploring what it means for robots to cohabit with humans in a traditional Korean home — a Hanok (한옥). * *But you may also choose to explore other non-apartment home types in Korea (e.g., countryside houses, villas, or sharehouses).* Your job is to create a **catalog page that shows what happens when futuristic domestic robots enter spaces shaped by history, culture, and ritual**.

You'll sketch, collage, and build your layout using the materials provided.

🧒 Focus on the ***"shared life* between robots and humans in these traditional homes."** How does the place itself aCect how robots behave, move, or relate to people?

✅ Your Design Brief

Create a catalog page that includes:

- **Hanok Scene / Background**
  - Show Hanok features (e.g., ondol, wooden doors, courtyard)
  - Where are the robots and humans? How do they navigate this traditional space?
  - What cultural values or rituals are preserved, altered, or lost?
- **Robots (at least 1)**
  - Give them names, prices, and short descriptions
  - What roles do they take in this setting?
  - How do they move and behave in this environment?
- **Users (Humans)**
  - Who lives in this house? (e.g., multigenerational family, solo elder, MZ generations?)
  - How do they interact with the robots?
- **Catalog Format**
  - Make it look like a real catalog page
  - Consider layout, aesthetic style, and text-image balance

💡 Hints & Things to Think About

- **Ideas from Workshop 1 - You can consider using/applying them on your page**
  - Robot types: house chore, family mediator, smart home manager... and many more
  - Cultural values and traditions ideas
- **More questions you can think about:**
  - Will this house change to accommodate robots (robot-friendly) or will the robots change to adapt to Hanok life?
  - How is this space diCerent from a modern apartment? What new challenges or opportunities arise? (Compare & Contrast your ideas with other groups.)

[5] Study/Workroom → *Designing a Future of Learning, Working, and Thinking with Robots*

Theme: 💼 *Designing Everyday Life (Work or Study) Look Like with Robots at Home?*

🧠 Your Mission

You are *Future HRI designers*, exploring what happens when robots become part of our daily work and study routines at home. Your mission is to create a **catalog page that captures scenes from a study or workroom** in a future Korean household. Imagine students writing papers, children learning math, or remote workers on Zoom calls with robotic companions present. What kinds of robots exist in this space and what roles do they take?

You'll sketch, collage, and create a layout using the materials provided.

🧒 For this mission, focus on **"future daily life with robots"**, not just how the robots look. Consider how they influence motivation, focus, productivity, or even mental health.

✅ Your Design Brief

Create a catalog page that includes:

- **Study / Workroom Scene / Background**
  - What's in the room? (desks, digital tools, charging stations, lighting, etc.)
  - What's the mood or vibe of this room?
  - How are humans and robots positioned within the space?
- **Robots (at least 2)**
  - Include names, prices, and short descriptions
  - What are their roles? (e.g., assistant, tutor, creative partner, motivator, timer..)
- **Users (Humans)**
  - Who's using this space? (e.g., student, freelancer, elder, child)
  - What are they doing?
  - How do they relate to the robots (e.g., supported, guided, controlled?)
- **Catalog Format**
  - Make it look like a real catalog page
  - Think about layout, design balance, and how to guide the viewer's eye

💡 Hints & Inspirations

- **Ideas from Workshop 1 - You can consider using/applying them on your page:**
  - Robot types: personal assistant... and many more
  - Ideas from the Post-it notes
- **More ideas to explore:**
  - Who is going to study (work) in the future? - *robots or humans?*
  - What if the study/workroom becomes a *"productivity panopticon"*?

[6] Bedroom → *Designing a Future Daily Life*

Theme: 🌙 *Everyday Nighttime Life with Robots*

💤 Your Mission

You've explored what happens when robots support us in daily tasks like work and study. Now, you're entering the most personal and private space of the home: the bedroom, a place of rest, healing, solitude, and intimacy. In this mission, you'll design a **catalog page that shows how robots interact with humans during bedtime routines, rest, or nighttime care**.

How do robots *fit into, or disrupt,* this quiet, vulnerable environment?

You'll sketch, collage, and create a layout using the materials provided.

🧒 Focus on "**how robots live *with* humans"** in this deeply personal space. Consider emotional needs, physical boundaries, or potential discomforts.

✅ Your Design Brief

- **Bedroom Scene / Background**
  - Show features of a bedroom (bed, lights, shelves, personal items)
  - What's the atmosphere of the room? (e.g., cozy, peaceful, clinical, tech-heavy?)
  - Where are the humans and robots? What's happening in the space?
- **Robots (at least 1)**
  - Give them names, prices, and short descriptions
  - What do they do? (e.g., sleep coach, cuddle bot, mood/health monitor?)
  - How do they behave diCerently in this quiet, vulnerable environment?
- **Users (Humans)**
  - Who is here? (e.g., single adult, child, couple, elder)
  - What support do they need from robots? Are robots trusted?
  - What are they doing? What are they feeling? How do they interact with the robots?
- **Catalog Format**
  - Think about the layout, typography, and placement to guide the reader's feelings

💡 Hints & Things to Think About

- **Ideas from Workshop 1 - You can consider using/applying them on your page**
  - Robot types: emotional support robot, nursing robot... and many more
  - Ideas from the Post-it notes
- **More questions you can think about:**
  - Is there a point where the robot becomes *too close?*
  - Who controls or programs this robot? How are the privacy issues handled?

[7] Shower / Laundry / Balcony / Doorstep → *Designing a Future of Shared Everyday Spaces*

Theme: 🚪 *What Happens When Robots Live in the Margins of the Home?*

🏠 Your Mission

You are *Domestic HRI Specialists*, exploring how robots exist in the often-overlooked but essential parts of daily life: the **shower, laundry room, balcony, or doorstep.** This is your chance to explore how robots adapt to our daily life. How do they fit in, assist, or reshape these places?

You'll sketch, collage, and build your layout using the materials provided.

🧒 Focus on **"shared life between humans and robots"** in spaces that are functional, transitional, or private. Think about privacy, hygiene, emotional boundaries, and the *design of cohabitation.*

✅ Your Design Brief

Create a catalog page that includes:

- **Scene / Background**
  - Choose one (or two) place(s): shower, laundry, balcony, or doorstep
  - What's the purpose of this space? What kinds of human activities happen here?
  - How does the space change with the presence of robots?
- **Robots (at least 1 for each place)**
  - Give them names, prices, and short descriptions
  - What do they do? (e.g., laundry bot, shower helper, home manager?)
- **Users (Humans)**
  - Who uses this space? (e.g., parent, child, solo adult, guest)
  - What are they doing? How do they feel in this moment?
  - How do they interact with the robots - *physically and mentally?*
- **Catalog Format**
  - Make it look like a real catalog page
  - Consider layout, visual hierarchy, and how you use text and image

💡 Hints & Inspirations

- **Ideas from Workshop 1 - You can consider using/applying them on your page:**
  - Robot types: shower robots, smart home managers...and many more
  - Ideas from the Post-it notes
- **More questions you can think about:**
  - Do you want to place a charging station somewhere? How will they be charged?
  - Who controls or programs this robot? How are the privacy issues handled?

[8] Entire Home → *Designing DiCerent Types of Robots for Future Home*

Theme: 🏘️ *What Kinds of Robots Can Be Adapted, Attached, or Embedded into the Home?*

🏠 Your Mission

You are now *Future Robot Designers*, imagining new kinds of robots for the homes of the future. Instead of focusing on one room or one moment, step back and view the home as an interconnected system, filled with spaces, emotions, routines, and technologies. What kinds of robots live here? Where do they belong? How do they move, attach, adapt, or blend in?

※ You can **move around**, **talk to other groups**, and even **come up with something strange, funny, or totally unrealistic**.

🧒 Focus on **"inventing diCerent types of robots that can exist in future homes."** Think about mobility, accessibility, attachment, visibility, and emotional presence.

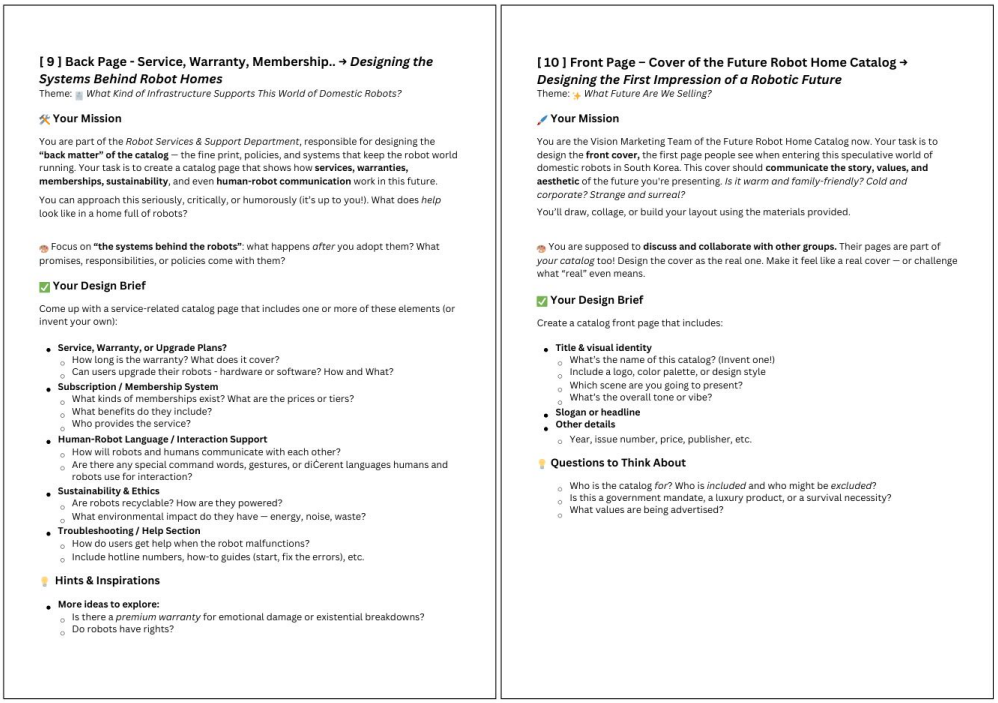
✅ Your Design Brief

Create a catalog page that includes:

- **Robots (at least 3)**
  - Give each robot a name, price, and functions
  - What is their role? Where are they (often) placed in the home?
  - Are they mobile, stationary, modular, wearable, or wall-mounted?
- **Catalog Format**
  - Make it look like a real catalog page
  - Think about layout, labeling, storytelling, and connections between robots and rooms
- **Things to think about:**
  - **Scene / Background**
    - How will these robots move and operate across the home?
  - **Users (Humans)**
    - Who is the main target of these robots?
    - How do they interact with the robots?

💡 Hints & Inspirations

- **Ideas from Workshop 1 - You can consider using/applying them on your page:**
  - Robot types: pet, smart home manager, robots from the movies... and many more
- **More ideas to explore:**
  - Will robot-robot interaction (RRI) happen? What would that look like?
  - Can a robot become a *legal family member* here?



[9] Back Page - Service, Warranty, Membership.. → *Designing the Systems Behind Robot Homes*

Theme: *What Kind of Infrastructure Supports This World of Domestic Robots?*

Your Mission

You are part of the *Robot Services & Support Department*, responsible for designing the **"back matter" of the catalog** — the fine print, policies, and systems that keep the robot world running. Your task is to create a catalog page that shows how **services, warranties, memberships, sustainability**, and even **human-robot communication** work in this future.

You can approach this seriously, critically, or humorously (it's up to you!). What does *help* look like in a home full of robots?

Focus on **"the systems behind the robots"**: what happens *after* you adopt them? What promises, responsibilities, or policies come with them?

Your Design Brief

Come up with a service-related catalog page that includes one or more of these elements (or invent your own):

- **Service, Warranty, or Upgrade Plans?**
  - How long is the warranty? What does it cover?
  - Can users upgrade their robots - hardware or software? How and What?
- **Subscription / Membership System**
  - What kinds of memberships exist? What are the prices or tiers?
  - What benefits do they include?
  - Who provides the service?
- **Human-Robot Language / Interaction Support**
  - How will robots and humans communicate with each other?
  - Are there any special command words, gestures, or diCerent languages humans and robots use for interaction?
- **Sustainability & Ethics**
  - Are robots recyclable? How are they powered?
  - What environmental impact do they have — energy, noise, waste?
- **Troubleshooting / Help Section**
  - How do users get help when the robot malfunctions?
  - Include hotline numbers, how-to guides (start, fix the errors), etc.

Hints & Inspirations

- **More ideas to explore:**
  - Is there a *premium warranty* for emotional damage or existential breakdowns?
  - Do robots have rights?

[10] Front Page – Cover of the Future Robot Home Catalog → *Designing the First Impression of a Robotic Future*

Theme: *What Future Are We Selling?*

Your Mission

You are the Vision Marketing Team of the Future Robot Home Catalog now. Your task is to design the **front cover,** the first page people see when entering this speculative world of domestic robots in South Korea. This cover should **communicate the story, values, and aesthetic** of the future you're presenting. *Is it warm and family-friendly? Cold and corporate? Strange and surreal?*

You'll draw, collage, or build your layout using the materials provided.

You are supposed to **discuss and collaborate with other groups.** Their pages are part of *your catalog* too! Design the cover as the real one. Make it feel like a real cover — or challenge what "real" even means.

Your Design Brief

Create a catalog front page that includes:

- **Title & visual identity**
  - What's the name of this catalog? (Invent one!)
  - Include a logo, color palette, or design style
  - Which scene are you going to present?
  - What's the overall tone or vibe?
- **Slogan or headline**
- **Other details**
  - Year, issue number, price, publisher, etc.

Questions to Think About

- Who is the catalog *for*? Who is *included* and who might be *excluded*?
- Is this a government mandate, a luxury product, or a survival necessity?
- What values are being advertised?

A.4 Workshop Post-Questionnaire

[Post-questionnaire: Reimagining Human-Robot Future Homes in South Korea Using Speculative Design 🤖🏠]

Thank you so much for participating in the workshop! ☺️

Your reflections are incredibly valuable for my research on how Human-Robot Interaction (HRI) can reshape domestic spaces in South Korea through speculative design.

This short questionnaire is completely voluntary, but I would be truly grateful if you could take a few minutes to share your thoughts. (You are free to answer only some questions you want to answer and leave others blank.)

💡 You may submit your response by next Wednesday (April 9)
🗣️ English is preferred, but feel free to respond in Korean if that's more comfortable.
🔒 All answers will remain anonymous.

If you have any questions, feel free to contact me:
👤 Minjae Kim | ✉️ minjae.kim.6960@student.uu.se

- Post-Study Questionnaire –

1-1. Which workshop(s) did you participate in? (Select one or both) *

[ ] Workshop 1 (March 19th - Affective Science class)
[ ] Workshop 2 (March 26th - Journal Club)

1-2. Which session(s) of the workshop did you enjoy the most? (Check all that apply)*
[ ] 1. Introduction (Workshop 1&2 - about Speculative design, Human-Robot Interaction, etc.)
[ ] 2. Scenario Discussion (Workshop 1 - Brainstorming session using Post-it notes)
[ ] 3. Future/World Building (Workshop 1 - Individual Reflection using the design canvas)
[ ] 4. Catalog Design (Workshop 2 - Design Catalog page using ideas from the first workshop and other materials)
[ ] 5. Presentation/Discussion (Workshop 2 - Sharing ideas and design with each other)
[ ] I don't know
[ ] Other: _____

1-3. What did you enjoy most, and why? Is there anything you're taking away from the experience (big or small)?
1-4. Was there anything that felt unclear or less engaging during the workshop(s)? Or what could be improved?
2-1. After participating in the workshop(s), what kinds of futures — good or bad — did you imagine around robots and home life?

2-2. After participating in the workshop(s), do you think robots (or Human-Robot Interaction) can change or challenge social norms? Why or why not?

3. Is there anything else you'd like to share about your experience or the ideas? (e.g., overall workshop experience, reflection about the method, anything you didn't get a chance
to express during the session)

4. Please leave your email address if you want to receive any updates about this project (e.g., final paper, catalog, etc.).
You can send me another email if you want to leave the answer completely anonymous, but still want to receive updates.

A.5 Workshop Structures

Speculative HRI Design Workshop: Reimagining Human-Robot Future Homes in South Korea
[Workshop 1]
Structure
Date: March 19th, 2025
Duration: 1 hour (16:30 – 17:30), Guest lecture (workshop) during the Affective Science class
Location: Seoul National University Asia Center
Participants: Graduate school students, including members from the SNU Human Factors Psychology Lab
Number of participants: 14 people (maximum 15)
Language: English
Supplies: (TV) Screen, Paper, Pen, Post-it, Design Canvas

Plan

1. Preparation
   - Setting up the classroom
   - Participants complete the pre-questionnaire (demographics, prior knowledge, and experience) and sign the consent form

2. Introduction (15min)
   - Welcome participants and introduce the facilitator and thesis project
   - Explain the agenda and goals of the workshop
   - Give a short lecture introducing "Speculative Design" – What it is, how it differs from traditional or critical design, and examples of existing speculative design artifacts
   - Introduce Speculative Design within Human-Robot Interaction (HRI) - explaining what HRI is and how speculative design can be applied in this field
   - Present the speculative "what-if" scenario: *"What-if all families have to adopt at least one robot in their home?"*

3. Scenario Discussion (30min)
   - Conduct rapid-fire brainstorming using Post-it notes around different related topics
   - Explain the rules and reminders to encourage active participation:
     - All participants stand up, pick a pen and notes
     - Write one idea per post-it, say it out loud, and stick it on the board
     - Spend about 2 – 5 minutes per topic
     - Emphasize "There are no wrong answers."
     - Focus on quantity over quality
   - Discussion topics:
     - How do you want to adopt robots in your home? (broad, basic, easier to answer) (3min)
     - What will they do? / What do you want them to do? (role, feature, tasks) (3min)
     - What shouldn't they do? / What do you not want them to do? (3min)
     - Think about the possible positive future for society & domestic places with robots (4min)
     - What unexpected problems might arise when robots are in your home? (4min)
     - What & How do cultural traditions or values change? (3min + 5min)

4. Future/World Building (15min)
   - Each participant works individually on a design canvas (A4 paper), with discussion allowed
   - Expand upon rapid-fire brainstorming ideas by filling in the design canvas:
     - Technology – Name/Definition, Features/Tasks (3-5)
     - Sketch – Design (2min max for drawing), Personality
     - Context – Scenarios (Human-Robot, Human-Human, Robot-Robot interaction), User/Target/Persona, Place/Location
     - Time – Macro Developments (world now -> world then)
     - Spark – Questions/Visions/Random ideas
     - Optional – Draft a short guide or advertisement for the robot

- Design one robot, but not only think about its features, but think about the surrounding world - how the world/society/different types of interactions will be changed

5. Outro
   - Thank the participant and invite them to join the next workshop
   - Collect all materials and clean up the room

[Workshop 2]

Structure

Date: March 26th, 2025
Duration: 2 hours (10:30 – 12:30)
Location: Seoul National University College of Social Sciences Seminar Room
Participants: Graduate school students and researchers, primarily from the SNU Human Factors Psychology Lab
Number of participants: 10 people
Participants Composition:
(1) 9 from the Lab, 1 external participant from Workshop I
(2) 5 returning participants from Workshop 1, 5 newly joined participants
Language: English (instruction), Korean (discussion)
Supplies: (TV) Screen, Basic school supplies (pen, paper, post-it notes, scissors, glue, tape, etc), IKEA catalogs, Mission file x 5 including instruction/mission page and materials to design catalog prototypes (e.g., different kinds of robot pictures, filled-in design canvas, related post-it note ideas)

Plan

1. Preparation
   - Setting up the room
   - Participants complete the pre-questionnaire and sign the consent form (only newcomers)

2. Introduction (10min)
   - Welcome participants and reintroduce the facilitator and thesis project
   - Explain the agenda and goal of the workshop
   - Provide a brief review of "Speculative Design," Speculative Design & Human-Robot Interaction (HRI), and the speculative what-if scenario

3. Workshop I Review (5min)
   - Recap the activities and outcomes from Workshop I, including scenario discussion and future/world-building
   - Highlight key ideas and insights generated

4. Designing the speculative/future artifact (95min)
Introduction (10min)
   - Introduce the design task: creating a "Future Robot Home Catalog" prototype
   - Present reference - Near Future Laboratory & Mobile Life Centre's "Future IKEA catalog" [10]

- Explain how the activity will proceed:
    - IKEA catalog composition
    - 2 catalog pages per group and per session
    - Each session lasts approximately 30- 40 minutes, two sessions in total
    - Each group will be assigned two tasks/missions with a thematic connection
    - Hints, references, and photos are provided; artistic skill is not prioritized (Generative AI use is discouraged.)
- Group selection based on participant interests or personality – Group sorting messages
- Distribute Mission Folders (with instruction/mission page, related design canvas, and post-it ideas, IKEA catalog page samples, and pictures of different robots and houses)

- Groups & Missions:

No	Group Sorting	Location	Session 1	Session 2
1	*"I like to look at things from all angles — both the good, the bad, and the in-between."*	Living Room / Kitchen	Positive Future	Dystopian Future
2	*"I'm drawn to stories, traditions, and the feeling of something special."*	Living Room / Kitchen (not strictly designated)	Holiday Season	Traditional House – Different Types of House
3	*"I love noticing the small things in daily life — routines, habits, and emotions."*	Other Rooms	Study or Workroom	Bedroom
4	*"I notice design, layout, and packaging — I care about how things look and feel."*	Entire Home	Shower or Laundry	Product-Based Pages – Different Kinds of Robots
5	*"I enjoy figuring out how things work — manuals, rules, fine print don't scare me."*	Front, back and Service page	Service guide / Guarantee / Membership / Sustainability	~~Front Page~~ -> *dropped due to logistics*

Session 1 (40min)

- Each group works on the first mission
- Short group presentations (~2 min for each) explaining the mission, outcomes, and reflections

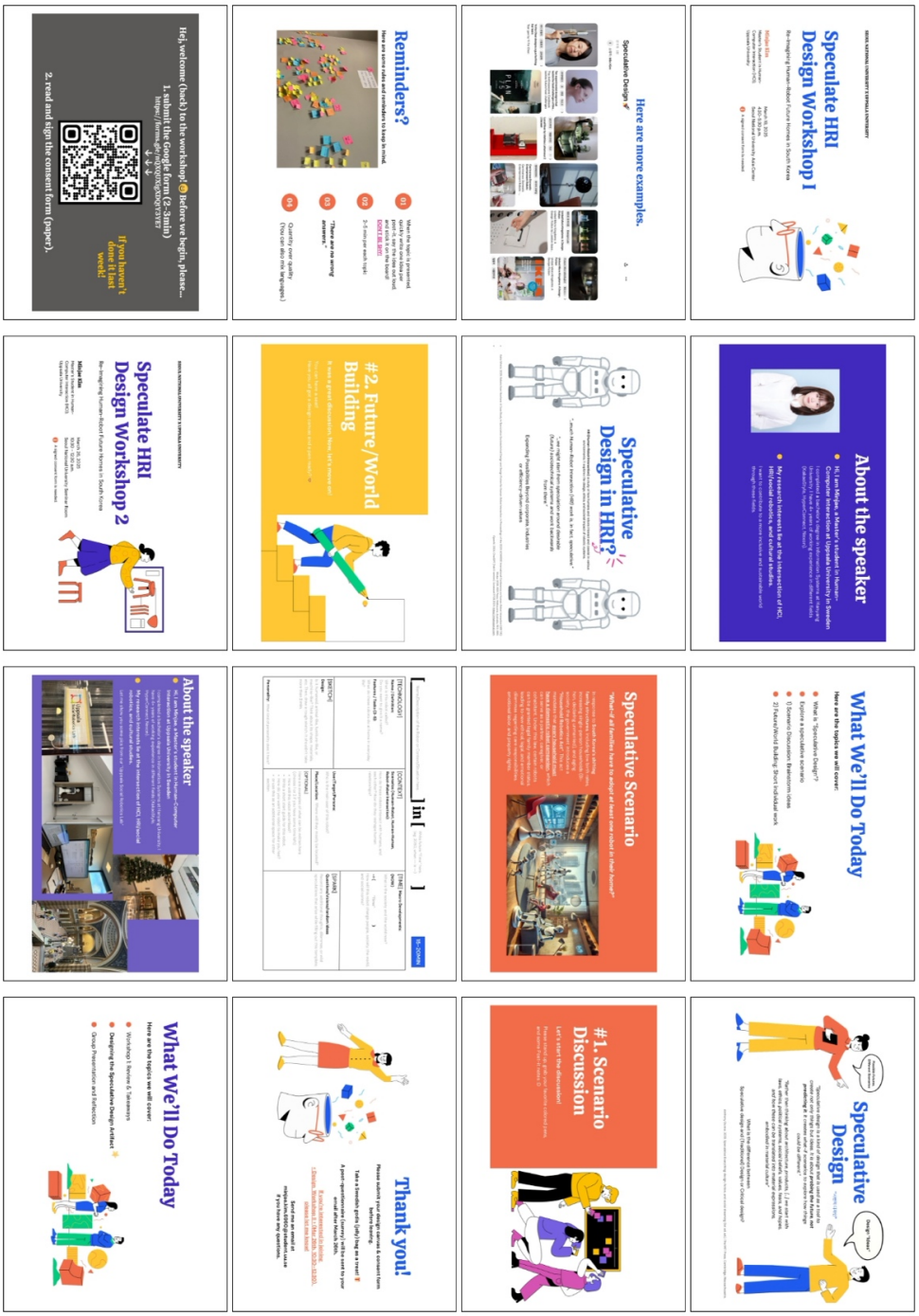
Break (5min)

Session 2 (40min)

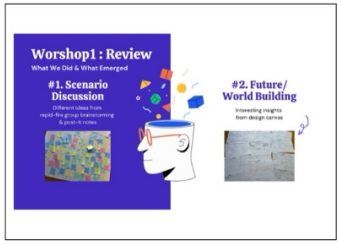
- Groups proceed to their second mission
- Short group presentations (~2 min for each) explaining the mission, outcomes, and reflections
- Intermediate brief interview with groups that finish early

5. Outro (10min)
    - Open floor for participants to share reflections and takeaways on the workshop experience, speculative design, and possible future works
    - Thank the participants
    - Collect all materials and clean the room

A.6 Workshop Slides (Link to Page)

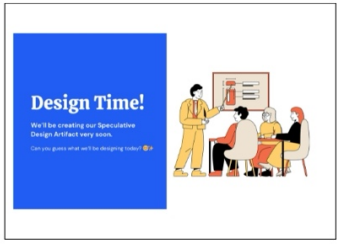



Speculative Design & HRI


Worshop1 : Review
What We Did & What Emerged
#1. Scenario Discussion
#2. Future/ World Building


Key Topics and Ideas
Wanted
Not-wanted
Positive future
Unexpected problems
Cultural traditions and values


Insights


Design Time!
We'll be creating our Speculative Design Artifact very soon.

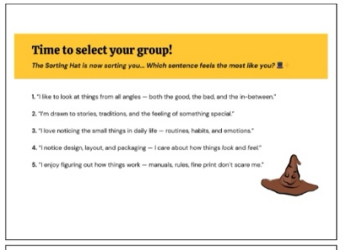

"Future Robot Home Catalog"
IKEA


Please, don't be afraid!

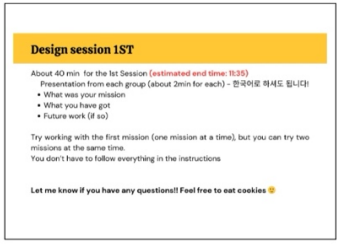

Quick Look
What does the IKEA catalog look like?
IKEA

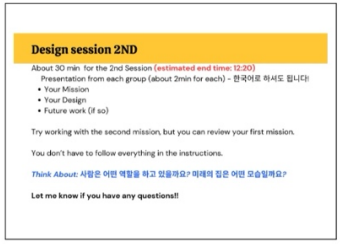


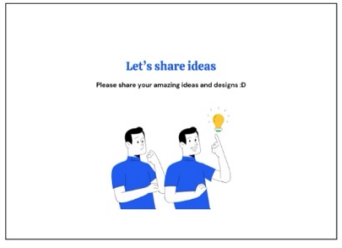

Be Prepared
01
Two+ pages per one group & session
02
"You are all designers."
03
Have fun!

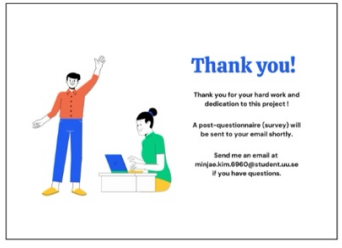

Time to select your group!
The Sorting Hat is now sorting you.. Which sentence feels the most like you?
1. "I like to look at things from all angles — both the good, the bad, and the in-between."
2. "I'm drawn to stories, traditions, and the feeling of something special."
3. "I love noticing the small things in daily life — routines, habits, and emotions."
4. "I notice design, layout, and packaging — I care about how things look and feel."
5. "I enjoy figuring out how things work — manuals, rules, fine print don't scare me."


Here is your mission folder


Design session 1ST
About 40 min for the 1st Session (estimated end time: 11:35)
Presentation from each group (about 2min for each) - 한국어로 하셔도 됩니다!
• What was your mission
• What you have got
• Future work (if so)
Try working with the first mission (one mission at a time), but you can try two missions at the same time.
You don't have to follow everything in the instructions
Let me know if you have any questions!! Feel free to eat cookies


Design session 2ND
About 30 min for the 2nd Session (estimated end time: 12:20)
Presentation from each group (about 2min for each) - 한국어로 하셔도 됩니다!
• Your Mission
• Your Design
• Future work (if so)
Try working with the second mission, but you can review your first mission.
You don't have to follow everything in the instructions.
Think About: 사람은 어떤 역할을 하고 있을까요? 미래의 집은 어떤 모습일까요?
Let me know if you have any questions!!


Let's share ideas
Please share your amazing ideas and designs :D


Thank you!
Thank you for your hard work and dedication to this project !
A post-questionnaire (survey) will be sent to your email shortly.
Send me an email at minjae.kim.6960@student.uu.se if you have questions.

A.7 Workshop Consent Form

[Participant Consent Form]

Study Title: Reimagining Human-Robot Future Homes in South Korea Using Speculative Design
Researcher: Minjae Kim (Uppsala University, Department of Informatics and Media)
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Introduction: You are invited to participate in a research study exploring human-robot interaction (HRI) and its implications for future domestic spaces. Before deciding to participate, please read this consent form carefully. It explains the purpose of the study, what your participation involves, and your rights as a participant.
Purpose of the Study: The purpose of this study is to explore how human-robot interaction can reconfigure future domestic spaces through speculative design. Your participation will help generate insights into designing future robots for domestic environments in South Korea.

What Will You Be Asked to Do?

- Participate in a workshop (which will take approximately 1-3 hours over 1-2 sessions) consisting of discussions and design activities.
- With your permission, the workshop may be audio/photo/video recorded for research purposes. These recordings will be used solely for data analysis, and no identifying details (i.e., face or voice) will be included in any final products.

Voluntary Participation & Right to Withdraw: Your participation is entirely voluntary. You may decline to answer any question or withdraw from the study at any time without any consequences. If you withdraw, any data collected from you will be deleted upon request.

Confidentiality & Data Protection

- Your personal information will remain confidential.
- Any data used in publications or presentations will be anonymized (i.e., no identifying details will be included).
- Workshop recordings, notes, and responses will be securely stored and only used for research purposes and will be stored only until the study is finished (approximately June 2025) and then securely deleted.

Consent & Agreement

Please check the following:

□ I have read and understood the information provided and agree to participate.
□ I consent to audio/photo/video recording during the workshop.

Participant's Name: _____
Signature: _____
Date: _____

A.8 Data Overview 1: All Brainstorming Post-it Notes (Grouped into Sub-themes)

Table 1: All Post-it Ideas Related to Sociotechnical Imaginaries (Desirable and Undesirable Robots and Futures)

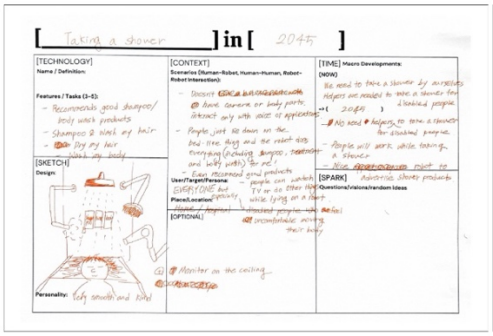
Prompt	Sub-theme(s)	Post-it Notes
Desirable Robots	Household chores	Automate chores, laundry, cleaning, cooking
	Private curator	Recommend clothes, meals (menu), movies, hairstyle, control diet, personal trainer (e.g. exercise advice)
	Personal assistant	Wake me up and help setting up, check daily routines and health, forecast weather, tell the news, shampoo and dry hair
	Smart working partner	Creative ideas, teach language, co-writing partner (emails, thesis, etc.), help finishing the assignment
	Emotional supporter	Ask about my day, listening do my daily episodes, consult about emotional state, empathy, emotional, have charming behavior (called Aegyo)
Undesirable Robots	Harmful behavior	Physical harm, attack humans, break stuff, being racist, linguistic violence, sexual harassment
	Malfunction or untrustworthy	Running out of battery when needed, not working, system errors, doing things that I have never asked, tell lies
	Manipulation and control	Recommend or encourage dangerous places/food/activities, manipulate feelings, decide on my behalf, spy on me
	Inappropriate behavior	Keep asking charging, begging me when I want to turn it down, be jealous of my guests, obey all my requests
	High cost	Too expensive to afford
Desirable Future	Efficient time use	Be PRODUCTIVE, efficient life (robots doing all the house chores), working everywhere & everytime, more time for hobbies, do only what I want to do
	Less loneliness	Less depression, mitigate social loneliness and isolation, foster responsibility (taking care of robots), pets that live forever with me, appreciate human contact, becoming friends, prevent dying alone
	Promoted health and well-being	Recommend healthy habits, increased public hygiene, always clean home, increased life quality, prevent physical and mental diseases
	Advanced technology benefits	Convenient life, makes me smarter, developed a legal system related to robots and human society
Undesirable Future	Negative social and environmental consequences	Social polarization, information gap, world-ending violence, loss of jobs, increased use of electricity, risk of privacy invasion
	Reduced (human) interaction	No marriage, forgetting the value of "life", forget social manners, avoid facing real humans
	Emerged ethical concerns	Robot rights, uprising, hierarchies, government controls
	Cognitive implications of technology dependency	Human intelligence decrease, too dependent, learn to do fewer things, get dumber, follows robots' recommendations (or norms) without thinking

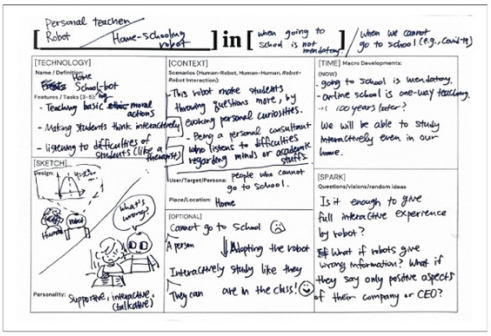
Table 2: All Post-it Ideas Related to Cultural Values and Societal Shifts

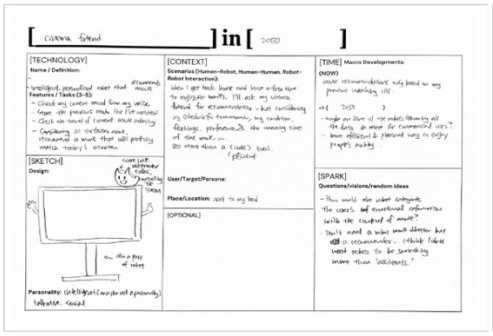
Sub-theme(s)	Cultural Value	Related Societal Changes
Culture and lifestyle	House (traditional house (Hanok), living in apartment, living in the city)	Robot-friendly interior, robot charging room, additional space for robots, traditional house -> smart high-tech house (IoT)
	Holiday (New Year's Day (Seollal), Chuseok, memorial service for ancestors)	Robot cook for holidays, virtual reunion

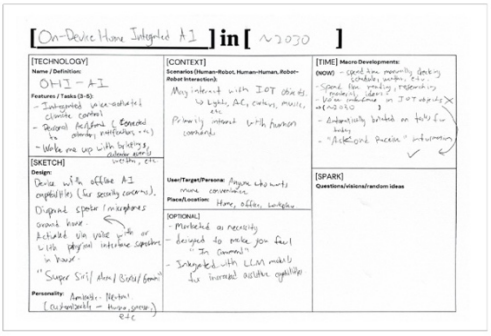
Sub-theme(s)	Cultural Value	Related Societal Changes
	Trends (food, art, brand, clothing)	People prefer the product robot recommends, new recipes, robot clothing, more standardized culture
	Education (high emphasis on education)	New types of specialized schools come out, robot-oriented education
Family dynamics and relationship	Parent-Children relationship (filial duty, rod of love)	Children obey robots, not their parents, baby's first word "Alexa"
	Family "roles" & appellations (extended family)	New definition of family, linguistic diversity in family
	Marriage	Need of finding life companion decreases, individual living
Social norms	Language (honorifics / polite words, formal-informal difference)	New human-robot language appears, changed into a way more easier for robots to understand, less using honorifics, only casual words teaching
	Labor (exchange of labor, hierarchies in company)	Online (hybrid) working, people use more commands to each other
	Respect, politeness, ethics, and beliefs	No sacrifice, less gathering and communication, individualism

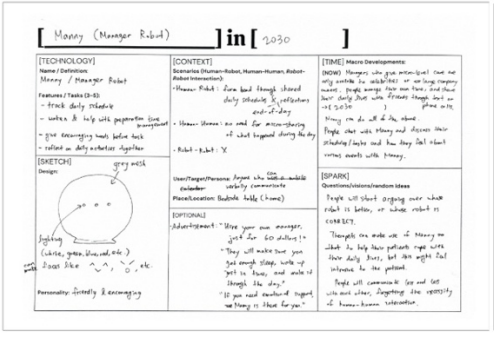
A.9 Participants' Design Canvas

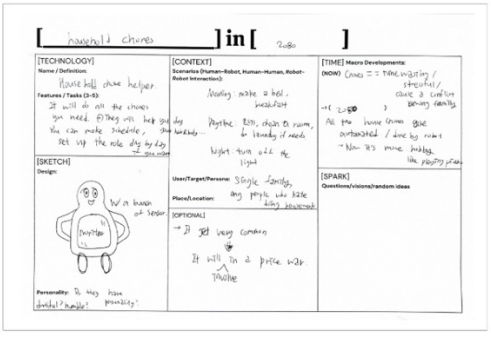


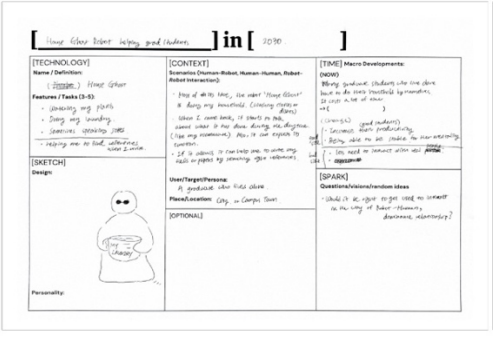


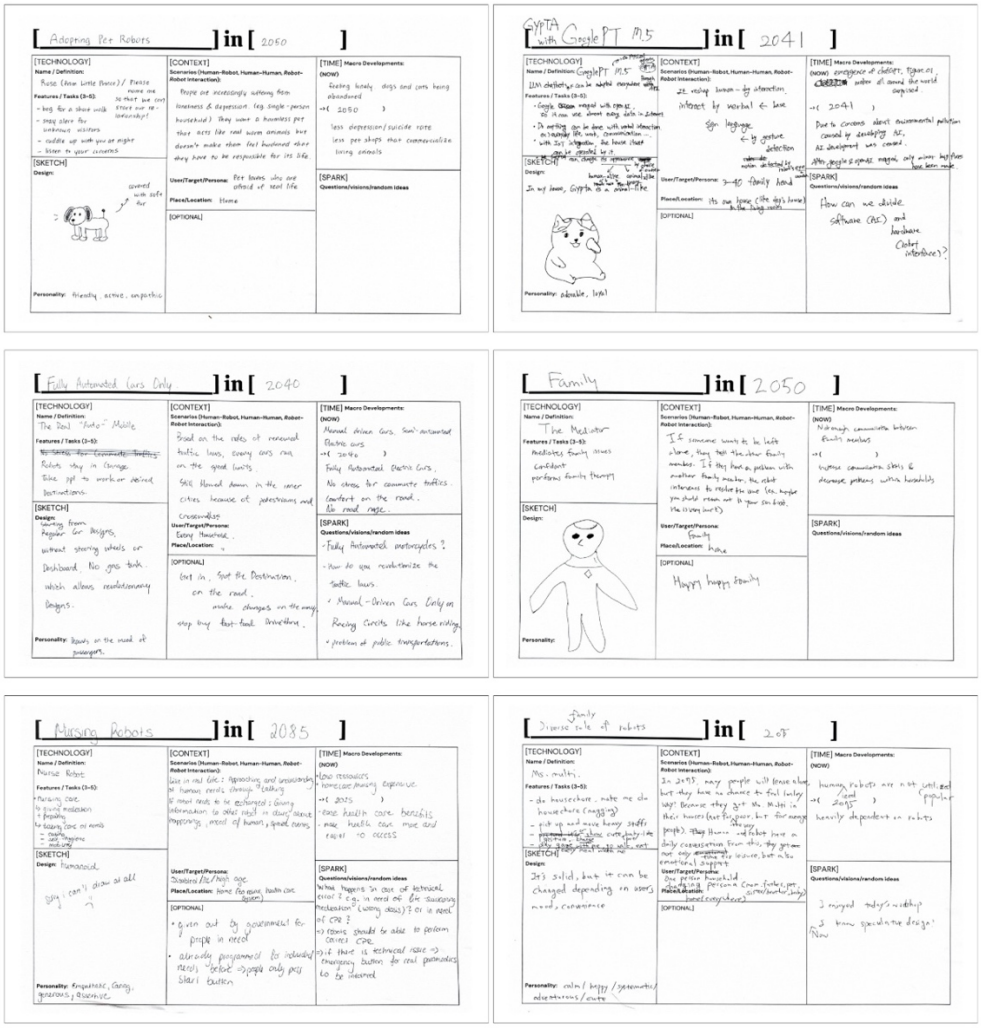












A.10 Data Overview 2 – Design Canvas: Speculative Robots, Interactions, and Worldbuilding Scenarios

ID	Robot	Interaction	World	Other(s)*
P1	Shower Bot - Automated shampoo, washing, and drying	- Interact only with voice or applications (no camera) - Robot washing while people lie down	- People will work while taking a shower - No need for helpers for disabled people to take a shower	
P2	Home Schooling-bot - Teach basing moral actions - Teach interactively - Work as a therapist	- Robots evoke students' curiosity - Listens to difficulties regarding mental and academic stuff	- Going to school is not mandatory - People cannot go to school (e.g. Covid-19)	*Q1. What if robots give wrong information?* *Q2. What if they say only positive aspects of their company or CEO?*

ID	Robot	Interaction	World	Other(s)*
P3	Cinema Friend - Intelligent robot that recommends a movie - Check current mode, adapted to a monitor	- Mainly recommends a movie, considering schedule, preferences, and mood	- Robot learning all the data in movies for commercial use - People enjoying hobbies more efficiently and pleasantly	R1. I think I don't want robots to be something more than 'assistant.'
P4	OHI-AI (On-device Home Integrated AI) - Climate control - Personal assistant	- Interact with voice (speaker and microphones around the house) or physical interface	- No need to check schedule manually or daily	V1. Robots marketed as necessities
P5	Manager Robot (Manny) - Track daily schedule - Time management - Emotional support	- Human-robot bond through shared schedule and daily reflections - People hire their manager robot	- Private manager (not only for celebrities) - Therapists use Manny to help their patient	V2. People will argue over whose robot is better or more correct *Q3. Will people feel intrusive?*
P6	Helper - Do all the household chores - Help you dry and set up	- Interact all day from morning to night - Interact by sensors and monitor	- All the house chores got automated by a robot - People having more time for hobbies	
P7	House Ghost - watering plants - doing laundry - speak jokes	- Robots start talking about what they have done during the daytime (like roommate) - Robots express emotions	- Increased human productivity - Less need to interact with real people	*Q4. Would it be right to get used to human dominance relationship with robots?*
P8	Darling - Household chores (cooking, cleaning, care home) - Humanoid robots can be customed to my type	- Robots do whatever (and whenever) humans want them to - Single households feel less lonely	- Less communication - Increased social polarization (on whether they own a robot)	V3. Robot adoption maybe not the best way to maintain society since people's desire to have a family will be decreased
P9	Pet Robot (Rose) - Basic pet dog features - Listen to concerns	- Harmless pets that act like real warm animals - Don't make people burdened	- Less depression and suicide rate - Less commercialized pet shops	
P10	GypTA (adopted everywhere) - LLM chatbot robots with animal look - IoT integration - Can change its appearance	- Reshape human-pet interaction - Interact by verbal, sign language, and motion detection	- AI development has ceased due to environmental concerns	*Q5. Can we divide software (AI) and hardware (robot) separately?*
P11	The real "Auto-" mobile - Robots staying in the garage - Driving - Regular card designs without wheels, gas tank, and dashboard	- Voice interaction	- No stress for commute - Manual-driven, semi-automated electric cars -> fully automated electric cars	V4. New laws developed specifically for newly developed robots
P12	The Mediator - Mediate family conflicts - Performs family therapy	- Solves family conflicts - Robots intervene to resolve the issue	- Increased communication between family members - Decreased household problems	
P13	Nurse Robot - Nursing care, giving medication	- Understand human needs through talking	- Easier access to health care (price, timewise)	V5. Robots distributed to the public by the government

ID	Robot	Interaction	World	Other(s)*
	- Taking care of needs - Humanoid robot	- Robots give information to other robots and communicate		*Q6. What happens in case of technical error?*
P14	Ms-Multi - Do household chores - Have cute attitudes - Daily activities with people	- People feel less lonely - People spend time with robots for leisure and emotional support	- People become heavily dependent on robots	

* : Q=Question, V=Vision, R=Random ideas

A.11 Data Overview 3 – Catalog Design Prototypes: Speculative Robots, User Interactions, and Domestic Scenes

Group	Theme / Scene	Robot(s) Featured	Domestic Interactions	Notes / Captions
G1 – 1	Positive future	1. Yana Bot: Humanoid all-mighty house chore robot (home care, massage), emotional companion 2. CatBot: Customizable emotional companion	- A person relaxing comfortably on the couch - Individuals receiving massages and nursing care from a robot - A child playing with a cat-like robot	-
G1 - 2	Dystopian future	Same robots as G1-1	- A person lying passively on the couch, being lazy and bord - Mothers no longer engage in caregiving, as robots handle all household tasks - A child lacking communication skills due to limited human interaction and reliance on robots	N1. Same page design as G1 -1, but with warning notes
G2 - 1	Holiday season page	1. Ddak-bot: Humanoid robot, holiday food preparation, childcare 2. Projector-bot: Helping virtual reunion with faraway relatives	- A family having a virtual reunion with grandparents - A family celebrating holidays while a robot prepares holiday meals - Children playing with robots	*C1. Perfect preparation for the Lunar New Year holidays!* *C2. A relaxing holiday free from the holiday burden*
G2 - 2	Traditional house "Hanok"	1. Garden keeper: Specialized in Hanok gardening, incorporating gardening skills of ancestors 2. Insects eater: Pest control and disinfection 3. Sweet bot: Greeting guests	- Children playing in a garden and in traditional Hanok rooms - A person greeted and introduced to home by robots	C3. *Simply solve concerns related to Hanok!* C4. *Complete your wonderful house with robot friends* ☺
G3 - 1	Study or Workroom	1. PPOMO: Education assistant robot, monitor (surveillance) children's focus level 2. ARO: Work assistant robot, manages other tasks so people can focus on their work	- A child happily interacting with a small robot at a desk - An elderly woman sitting at her desk while a friendly-looking robot offers her a glass of juice	*C5. Concentrating only on work and study*

Group	Theme / Scene	Robot(s) Featured	Domestic Interactions	Notes / Captions
G3 - 2	Bedroom	1. JAMI (Korean word "sleep"): Sleep awareness cushion with IoT equipped, temperature control 2. COMI (Korean word "snore"): Bedroom pet robot showing media contents, communicate, alarm clock, , leave the beeroom for privacy when people fall asleep	- A couple sleeping in bed with the COMI robot positioned beside them - A man sleeping while holding the JAMI robot	*C6. Relaxation and recovery, Happy sleep*
G4 – 1	Shower or laundry	1. Air sauna: Hands-free showering, air sauna made with cypress tree, self-cleaning, aroma and mold prevention	(No user depicted in this page)	-
G4 – 2	Different kinds of robots	1. Rose: Dog-looking pet robot with fur options, emotional companion 2. DIO: Humanoid singing and dancing robot, plays with kids 3. Manager: Humanoid robot doing house chores	- An elderly woman holding a dog-shaped robot, appearing comforted and less lonely - Children plays with DIO - A couple having a conversation while drinking coffee prepared by a humanoid robot	*C7. Focus on time with your family* N2. Filial duty product special event to celebrate Parents' day with special discount
G5 - 1	Service guide page	(No robot depicted in this page)	(No user depicted in this page)	N3. 1 month trial service: automatic recommendation system, pop-up store experience, monthly subscription program N4. General service: regular inspection, warranty, 24/7 customer support